

RESEARCH REPORT SERIES
(*Survey Methodology rsm2026-03*)

**Cognitive and Usability Pretesting of the Adapted Self-Administered National
Telecommunications and Information Administration Internet Use Survey**

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Report issued: March 2026

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Abstract

The Current Population Survey (CPS), which is currently administered entirely as a computer assisted personal interview (CAPI) survey, is exploring adding a self-administered mode option to its data collection later this decade (Linse & Johnson, 2023). As a result, many supplemental surveys that are fielded as part of the CPS may also be impacted; one of these supplemental surveys is the National Telecommunications and Information Administration (NTIA) Internet Use Survey (also called the “CPS Computer and Internet Use Supplement”). In summer 2023, following cognitive testing of the CAPI instrument, the Center for Behavioral Science Methods (CBSM) at the U.S. Census Bureau conducted exploratory cognitive and usability testing of an adapted self-administered NTIA Internet Use Survey web instrument. There were a series of global changes incorporated in the self-administered questionnaire prior to testing, including modifying household-level questions where a participant had to report household members who engaged in specific Internet behaviors, changing the conversational language typically found in CAPI instruments, formatting the question-level instructions, and shortening the text in introductory sections. These global changes were made to reduce respondent burden and take advantage of the features of the Internet mode and were mostly successful in testing. We also evaluated select questions, to see if there were any changes in comprehension between CAPI and Internet. Overall, questions that participants did not struggle with in CAPI testing were also not difficult for them to answer in the Internet mode. Many questions that participants struggled with in CAPI testing also were difficult in the Internet mode; however, there was one question that was easier to answer in Internet. This exploratory cognitive and usability testing was a successful first look at how the question types and topic areas of the survey perform in a new mode.

Keywords: self-administered testing, instructions, question types, mark all that apply,

Suggested Citation: Jonathan Katz, Alexandra Piccirillo, Caitlyn Keeve, Jessica Holzberg, and Kristen Kohm. (2026). **Cognitive and Usability Pretesting of the Adapted Self-Administered National Telecommunications and Information Administration Internet Use Survey.** *Research and Methodology Directorate, Center for Behavioral Science Methods Research Report Series (Survey Methodology rsm2026-03)*. U.S. Census Bureau. Available online at <https://www.census.gov/library/working-papers/2026/adrm/rsm2026-03.pdf>

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EXECUTIVE SUMMARY

In an interagency project between the National Telecommunications and Information Administration (NTIA), under the Department of Commerce, and the U.S. Census Bureau, the Center for Behavioral Science Methods (CBSM) conducted cognitive interview and usability research on the NTIA Internet Use Survey. The NTIA Internet Use Survey is a computer assisted personal interview (CAPI) survey that is a supplement to the Current Population Survey (CPS). In preparation for the November 2023 fielding of the CAPI survey, CBSM conducted cognitive testing of the current CAPI instrument (see Katz, Piccirillo, Keeve, & Holzberg, 2024).

CBSM also conducted exploratory testing to evaluate a self-administered instrument version of the survey. There is a current initiative to add a self-administered component for the CPS, and later this decade, CPS supplements may also add a self-administered component. When developing the self-administered instrument, CBSM incorporated various changes, including but not limited to updating the format of the questions and instructions.. One round of cognitive and usability testing was conducted with 20 participants to evaluate the ramifications of these changes on data quality, participants' comprehension of the questions, and burden.

Participants were able to complete the adapted self-administered component of the NTIA Internet Use Survey without much issue and the changes that were incorporated were mostly successful. Select questions that were difficult in prior CAPI testing of the survey were also found to be difficult in the self-administered component. However, one question was found to be much easier to answer for participants in the self-administered component, the home Internet service question. Although testing of the adapted questionnaire was successful, future research should continue to evaluate how the CAPI and newly adapted self-administered survey perform. The research approach described in this report can also be a useful first step for other surveys exploring implementation of a self-administered mode.

1. INTRODUCTION

The Center for Behavioral Science Methods (CBSM) conducted cognitive and usability testing of an adapted self-administered version of the 2023 NTIA Internet Use Survey, a supplemental survey of the Current Population Survey (CPS). The CPS collects data on employment and labor work force but also conducts supplements on various topics in different months, such as the NTIA Internet Use Survey.

The NTIA Internet Use Survey, a computer assisted personal interview (CAPI) survey, has been administered periodically since 1994 and was last fielded in November of 2023. The survey asks questions about the household's use of devices that connect to the Internet, use of Internet from various locations in the past six months, different applications used to connect to the Internet, and any privacy or security concerns using the Internet. As part of data collection, one household member is randomly selected for additional questions about different Internet application uses. The household respondent may need to serve as a proxy respondent on behalf of the randomly selected household member during this section of the questionnaire.

The CPS is exploring a modernization effort that includes adding a self-administered component to its data collection (Linse & Johnson, 2023). In the years following, select supplemental surveys to the CPS will add a self-administered component as well. We wanted to evaluate whether participants would be able to complete a version of the NTIA Internet Use Survey adapted for the Internet. CBSM adapted the existing CAPI survey to a self-administered Internet mode prior to testing. This required changes to various survey questions for both practical reasons and also to maintain comparability in the question wording. After modifying the questionnaire, we conducted cognitive and usability testing of the adapted self-administered instrument.

2. BACKGROUND

Many computer-assisted personal interview (CAPI) surveys are starting to consider adding a self-administered web survey component to their data collection due to concerns about costs and rising nonresponse rates. The introduction of a new mode requires changes to the survey's contact strategy, questionnaire, interviewer training, processing, and statistical processes. In this section, we highlight some of the existing research focused on adapting a CAPI questionnaire to an Internet response mode.

Surveys that move from a CAPI design to a self-administered design should be visually appealing and easy to use (Betts & Cubbon, 2015). Further, web surveys in particular might be accessed on different platforms including smartphones, tablets, and computers. Therefore, the survey should display similarly across platforms (Dillman et al., 2014). Adapting a CAPI

instrument for a self-administered response also requires that some questions be reformatted to lessen respondent burden. For example, a list of Yes/No questions administered in an interviewer instrument can be modified to ‘Select-all-that-apply’ style questions in a self-administered instrument. However, this can increase satisficing as Callegaro et al. (2015) and Nicolaas et al. (2015) found there are more positive endorsements in a Yes/No format compared to ‘Select-all-that-apply’ format. For survey questions asking for a numeric or dollar amount, Dillman et al. (2014) suggest formatting the size of the response text box to fit the answer to the question (i.e., not providing an overly short or overly long text box). Further, providing other cues such as a dollar symbol next to the response box can guide respondents on how to answer the question without providing extra written instructions. Dillman and colleagues (2014) also indicate that including yes/no filter questions before asking a series of follow-up questions can lessen respondent burden. This can make skip instructions difficult for respondents in paper surveys, but in web surveys yes/no filters can ensure respondents do not need to answer questions that are not applicable to them.

The handling of interviewer instructions in self-administered instruments also needs to be considered. Some instructions give the respondent information about what to expect in the coming questions; this can be easily mirrored in a web-administered version but the instructions can be reduced (Campanelli, 2021).

Introducing a self-administered web survey raises concerns about data quality. Respondents may interpret questions differently depending on whether they read the question on screen or if an interviewer reads them the question. This can affect the comparability of estimates between self-administered and interviewer-administered data collections. A switch in data collection mode can have major impacts on data trends for topics that have been measured for decades. Jackle et al. (2015) found that item non-response increases in cases where surveys include a self-administered option; Cernat and Revilla (2020) also found higher item nonresponse in the web mode compared to CAPI. However, Loomis and Paterson (2018) found that the item nonresponse between CAPI and self-administered web was not significantly different. When “don’t know” is offered as a response option, it is more likely to be selected in self-administered web surveys than in face-to-face surveys (Lipps and Monsch, 2022; Wolf et al. 2021). Similarly, Breton et al. (2017) found “don’t know” and “refused” responses increased in web.

For this study, we wanted to explore the possible effects of a mode switch on the NTIA Internet Use Survey by cognitive and usability testing the adapted Internet version. This testing was conducted in summer 2023 and followed cognitive testing of the CAPI 2023 NTIA Internet Use Survey in spring 2023.

3. METHODOLOGY

3.1 Expert Review

Prior to testing, CBSM conducted a full review of the current NTIA Internet Use Survey CAPI instrument. CBSM drafted a recommended version of the self-administered instrument and then NTIA reviewed the questionnaire to provide feedback. NTIA accepted most of the recommended changes to the instrument to make it self-administered. These questionnaire changes were used to inform the focus of the cognitive and usability protocol.

Global changes made across the questionnaire for the Internet mode included:

1. Adapting select household level questions from two questions to one question.

In the CAPI survey, respondents are currently asked if anybody in the household currently uses specific devices connected to the Internet and at which locations they use the Internet. If it is a single-member household, the question is simply phrased as “Do you use...” For multi-member households, it is modified to “Does anyone in this household...” When respondents in multi-member households report ‘Yes’, they are then subsequently asked who in their household participates in each Internet activity. After respondents report the names of the household members that did each behavior, the interviewer marks each name retrieved from the roster completed earlier in the interview as applicable.

For self-administered testing, the wording for single-member households remained consistent with the CAPI mode. For multi-member households, we combined the two questions and modified the wording to be “Who in your household uses...” with each name auto-populated from the roster as a response option. Participants were instructed to mark the name of each person who did that activity. If no one participated in the activity, they were supposed to select “None of the above.” Therefore, participants only had to answer one question for each topic in Internet rather than the two questions in CAPI testing. Table 1 shows an example of these differences between CAPI wording and self-administered wording.

Table 1. Differences between CAPI and self-administered in laptop question

CAPI wording	Self-administered wording (multi-member household only) ¹
[Do you/Does anyone in this household, including you,] use a laptop computer at least occasionally? Include any laptop computers [you/members of your household] currently have access to and have used during the past six months. <i>(If needed)</i> A laptop is a personal computer that is portable and includes a built-in keyboard and screen. ☐ Yes ☐ No <i>(If yes & is multi-person household) Who is that?</i> <i>Mark all that apply.</i> <i>(Names not displayed to respondent)</i> ☐ Name 1 ☐ Name 2 ☐ Name 3	Who in your household uses a laptop computer at least occasionally? Include any laptop computers members of your household currently have access to and have used during the past six months. <i>A laptop computer is a personal computer that is portable and includes a built-in keyboard and screen.</i> <i>Select all that apply.</i> ☐ Name 1 ☐ Name 2 ☐ Name 3 ☐ None of the above

2. Changing conversational language that was specific to CAPI.

In the CAPI survey, many questions start with “What about...” to read more conversationally when asking about specific Internet behaviors after an initial question directing who the question is asking about (see Table 2).

¹ Question wording for single-member households can be found in Appendix B.

Table 2. Current CAPI Wording

CAPI wording
DESKTOP [Do you/Does anyone in this household, including you,] use a desktop computer at least occasionally? Include any desktop computers [you/members of your household] currently have access to and have used during the past six months. <i>(If needed)</i> A desktop is a personal computer that must be plugged into a power outlet and is typically at a single location. <i>(If yes & is multi-person household)</i> Who is that?
LAPTOP What about a laptop? [Do you/Does anyone in this household] use a laptop computer? <i>(If needed)</i> A laptop is a personal computer that is portable and includes a built-in keyboard and screen. <i>(If yes & is multi-person household)</i> Who is that?
TABLET What about a tablet or e-book reader, such as an iPad or Kindle? <i>(If needed)</i> A tablet or e-book reader is a personal computer that is portable and has a built-in screen, but does not have a built-in physical keyboard. <i>(If yes & is multi-person household)</i> Who is that?

For self-administered testing, all of these instances of “What about...” were removed and simply replaced with “[Do you/Does anyone in this household/Who in your household]....” Table 3 shows an example of the tablet and video watching questions that were modified to remove conversational text.

Table 3. Differences between CAPI and self-administered in tablet question

CAPI wording	Self-administered wording
TABLET What about a tablet or e-book reader, such as an iPad or Kindle? <i>(If needed)</i> A tablet or e-book reader is a personal computer that is portable and has a built-in screen, but does not have a built-in physical keyboard. <i>(If yes & is multi-person household)</i> Who is that? <i>(Names not displayed to respondent)</i> ○ Name 1 ○ Name 2 ○ Name 3 VIDEO What about streaming or downloading videos? [Do you/Does NAME] watch videos over the Internet? Examples include YouTube and Netflix. ○ Yes ○ No	TABLET Who uses a tablet or e-book reader, such as an iPad or Kindle? • <i>A tablet or e-book reader is a personal computer that is portable and has a built-in screen, but does not have a built-in physical keyboard.</i> • <i>Select all that apply.</i> ○ Name 1 ○ Name 2 ○ Name 3 ○ None of the above VIDEO [Do you/Does NAME] watch videos over the Internet? <i>Examples include streaming or downloading videos on YouTube and Netflix.</i> ○ Yes ○ No

3. Moving 'if needed' instruction text from the question stem to its own line in bullet format and putting it in italics

Most of the if needed text for individual questions was italicized and put in bullet format below the main question stem. Table 4 shows an example of the same tablet and video watching questions that were modified both in terms of removing conversational text and adapting the instruction text.

Table 4. Differences between CAPI and self-administered in tablet question

CAPI wording	Self-administered wording
TABLET What about a tablet or e-book reader, such as an iPad or Kindle? <i>(If needed)</i> A tablet or e-book reader is a personal computer that is portable and has a built-in screen, but does not have a built-in physical keyboard. <i>(If yes & is multi-person household)</i> Who is that? <i>(Names not displayed to respondent)</i> ☐ Name 1 ☐ Name 2 ☐ Name 3 VIDEO What about streaming or downloading videos? [Do you/Does NAME] watch videos over the Internet? Examples include YouTube and Netflix. ☐ Yes ☐ No	TABLET Who uses a tablet or e-book reader, such as an iPad or Kindle? • <i>A tablet or e-book reader is a personal computer that is portable and has a built-in screen, but does not have a built-in physical keyboard.</i> • <i>Select all that apply.</i> ☐ Name 1 ☐ Name 2 ☐ Name 3 ☐ None of the above VIDEO [Do you/Does NAME] watch videos over the Internet? • <i>Examples include streaming or downloading videos on YouTube and Netflix.</i> ☐ Yes ☐ No

4. Making decisions about questions per page.

When adapting this questionnaire, we typically displayed one question per screen. We sometimes displayed similar questions together, but each of these screens had no more than four questions. For example, the questions that asked whether a household member used email, text or instant messaging, or social networks were all displayed on the same screen. For the question that asked who in the household used a laptop computer was on it's own screen given the length of the question. Evaluating the number of questions per screen was not a focus of testing.

3.2 Data Collection

One round of cognitive and usability testing was conducted in July and August 2023, with recruitment beginning in June. All interviews were conducted remotely via Microsoft Teams. After the interview, participants were paid \$40 in cash via priority mail in appreciation for their time. Interviews were approximately 65 minutes to account for both a “tech check” at the

beginning of the interview and the time the interviewer needed to complete the protocol. Cognitive/usability testing and recruitment screening were conducted in English.

3.2.1 Recruitment Criteria and Strategy

Our efforts aimed to recruit respondents of varied demographic backgrounds in terms of household relationship, household type, sex, ethnicity, race, age, education, and income level who typically participate in the NTIA Internet Use Survey. In prior iterations of testing an interviewer-administered survey, we tried to recruit households with limited Internet use. Given that we were testing a self-administered web instrument, we did not explicitly recruit on Internet use.

Since the survey also asks about Internet use of the household, we also targeted both related and unrelated households to evaluate the feasibility of participants responding for other household members in both of these types of households. We thought it might be more difficult for respondents to report Internet use for people they were not related to due to knowledge issues. For this study, CBSM considered a “related household” to be one in which the members of the pair described their relationship as spouses, unmarried partners, or boyfriend or girlfriend; or if a participant lived with their parent, sibling, or child; grandparent or grandchild; or aunt, uncle, or cousin. CBSM considered an “unrelated household” to be one in which the members of the pair did not describe their relationship as ‘related’ as defined above. Examples include friends or roommates.

Participants were recruited from multiple sources, including recruits who had completed a screener from a previous testing project, participants who had filled out a Qualtrics self-administered screening instrument sent out via a Census Bureau advertisement, and participants who had called or emailed to be screened via telephone in response to the Census advertisement.

3.3 Demographic Characteristics

We recruited a variety of participants in this study in terms of their household relationship, ages, male/female, and education levels. Most of the households had at least two people or more and about half were related. There was a variety in the relationships between the participant and the randomly selected household member for the questions requiring proxy response.² The majority of participants were female. Participants also tended to be aged 50 or

² For the randomly selected household member questions, we either chose the participant to answer for themselves or to answer for someone else in the household. We wanted to measure both participant’s comprehension of the questions and then feasibility answering as a proxy, when possible. Specific household members were selected based on their relationship to the participant to ensure we had a variety in testing.

younger and have a Bachelor's degree or higher. The majority of participants were in households with an income of \$100,000 or less. See Table 5 for the participant characteristics and Table 6 for the relationship characteristics of multiple member households.

Table 5. Participant Characteristics

Participant Characteristics	Round 1
N	20
Household	
Single member	3
Multiple member	17
Sex	
Female	12
Male	8
Hispanic	
Yes	1
No	19
Race	
White	14
Black or African American	5
Asian	1
Age	
18-34	9
34-50	6
51-64	3
65 and over	2
Education	
Completed high school	1
Some college, no degree	4
Associate's degree (AA/AS)	1
Bachelor's degree (BA/BS)	8
Post-Bachelor's degree (for example, MA, MS, Ph. D, JD)	5
Other degree or studies	1
Income	
Less than \$15,000	3
\$15,000 to \$24,999	0
\$25,000 to \$49,999	0
\$50,000 to \$100,000	9
More than \$100,000	8

Table 6. Relationship Characteristics of Multiple Member Households

Relationship Characteristics	Round 1
Type of Relationship³	
Related	10
Unrelated	7
Relationship to Participant for Proxy Questions (EMAIL to HOMIOT)⁴	
Self ⁵	9
Husband/wife/spouse	3
Unmarried partner	2
Parent(s)	2
Children	1
Siblings	0
Roommates	3

3.4 Interviewer Staffing and Training

Cognitive interviews were conducted by CBSM interviewers who were trained and experienced in cognitive interviewing. Before testing, all interviewers were trained on administering the NTIA Internet Use Survey questionnaire with the cognitive and usability interview protocol.

3.5 The Cognitive Interview Protocol

The cognitive and usability interview protocol focused on the adapted CAPI NTIA Internet Use Survey questions (See Appendix A). Before beginning the interview, interviewers informed participants that the purpose of the interview was to assess how well the questions worked. Interviewers also informed participants of their rights, explaining that participation is voluntary, and that their responses were confidential.

Participants were also asked to think aloud as they answered the survey questions. Think aloud is used to understand a participant's mental process when they go through a task (Willis 2007). In this case, we wanted real-time feedback on how participants were interacting with the adapted web instrument.

At the beginning of the interview, participants completed a shortened version of the roster asked in the main CPS. Participants were first asked to list all household members and then report the demographic characteristics of each person. The demographic questions included sex, age, relationship to the participant, education, race, and ethnicity. Questions about the employment status of each person were also administered.

³ Single member households are not included in this count.

⁴ In 11 interviews, the participant answered as a proxy for someone else in their household. We selected various types of relationships for testing to evaluate the feasibility of answering as a proxy.

⁵ This includes three single person households and six multiple person households where the randomly selected member was the participant.

Following the roster, the participant proceeded to the NTIA Internet Use Survey (See Appendix B for questions administered to single-person households and Appendix C for questions administered to multi-person households). There were six sections in the questionnaire, which are outlined in Table 7.

Table 7. Description of Questions

Section	Description	Household or One Household Member Only
1	Household's use of Internet connected devices, such as laptops, tablets, and smart phones	Entire household
2	Where household members use Internet (school, work, public places)	Entire household
3	Internet service plans, home or mobile	Entire household
4	Online activities such as email, social media, gaming, and health	Household participant or proxy for another household member
5	Various challenges the household experiences using the Internet, such as privacy concerns and identity theft	Entire household
6	Reasons why no one uses the Internet at home and if household currently or ever had a cable or satellite TV subscription	Entire household

After completing the questionnaire, interviewers asked the participant retrospective probing questions to evaluate the usability of completing the web instrument and comprehension of the survey items. In retrospective probing, interviewers ask follow-up questions about a group of survey questions after the participant has answered them. This technique allows participants to complete the survey without stoppage. However, participants may have some difficulty recalling their earlier thoughts about a survey response (Willis, 2005). CBSM chose retrospective probing over concurrent probing (i.e., follow-up questions that are asked directly after a participant answers a survey question) to avoid biasing participants' answers later in the questionnaire. To reduce potential recall issues, participants were shown the questions again in a powerpoint as each of the questions came up in probing.

The probes asked questions related to the usability of the survey, such as ease/difficulty of (1) completing a survey on web generally and (2) selecting multiple household members for questions that asked who in the household participated in each specific Internet activity. Participants were also asked if it would have been easier to complete the survey on web or via

face-to-face/telephone. Participants were then administered probes targeting comprehension of the questions. The survey questions we probed about included questions that were either (1) problematic in CAPI cognitive testing (see Katz et al., 2024) or (2) had undergone substantive changes when modified from interviewer-administered that we thought could impact the comprehension of the question. In the middle of retrospective probing, participants were asked to answer an alternate version of the Home Internet service question (see Appendix A). Katz et al. (2024) found in CAPI cognitive testing that participants struggled with answering this question so NTIA recommended an alternate version to test.

Finally, interviewers asked participants a set of debriefing questions about how easy or difficult it was for the participant to respond to the survey and if participants thought other people might consider any of the questions sensitive. Participants were then asked to share anything else they had not yet had a chance to mention.

3.6 Interview Consent

After the initial screening but before the test interview, participants completed an online consent form. The form explained how the participant's answers and feedback would be used and highlighted the participant's rights as a participant and confidentiality. These rights include the freedom to decline to answer any question or to stop the interview at any point. At the beginning of each interview, interviewers reviewed the participant's rights and asked the participant to give oral consent to participate with video and audio recording. All participants consented to participate and be audio and video recorded.

4. RESULTS

We will first provide findings for the global changes outlined in Section 3.1 (Expert Review) and then we will discuss cognitive interview findings of the individually tested questions. The questions in the survey are divided into the following sections: Types of Computing Devices Used, Location of Internet Use, Type of Internet Service Used, Computing Applications and Services Used, and Challenges and Concerns with Internet Use.

4.1 Global changes to wording in the self-administered instrument

The following global changes, mentioned earlier, were made for the self-administered instrument:

- Adapting select household level questions from two questions to one question (e.g., selecting household member's name(s) for each Internet activity).
- Changing conversational language that was specific to CAPI.

- Moving ‘if needed’ instruction text from the question stem to its own line in bullet format and also changing the font to be italics.
- Making decisions about questions per page.

With these changes, participants generally were able to navigate and complete the self-administered instrument without much problem. At the beginning of retrospective probing, participants were asked if the survey was easy or difficult to complete. Almost all of the participants reported that the survey was easy to complete:

- “Everything was responsive and it didn’t halt in any place.”
- "The questions were simple. They weren't long winded. I pretty much got an idea on what goes on in my house to answer the questions. They didn't have multiple sentences and the survey really wasn't that long and that's what made it easy."

Regarding the global changes, combining the two-part household level questions into one question seemed to ease the user experience since there was less questions to answer.

Participants were asked how easy or difficult it was to select names for survey questions that displayed a list of names for who participated in which Internet activity. Most of the participants who were asked this question said this process was easy. A couple of participants noted that it was difficult knowing what other people in the household were doing at times. This seemed to be more of a knowledge issue than an issue with the question wording (which included a “Select all that apply” instruction) or display itself. Almost everyone seemed to find the process of selecting each name easy. Two participants noticed that a square button that was next to each name indicated to them it was a ‘Select all that apply’:

- “When I see square, I think check box. You can always check multiple options.”

Only one participant seemed to struggle with selecting names in the survey as it sometimes required them to make multiple clicks to mark the name. Although the participant did not mention this, we think it may have been a function of the participant’s mouse/touch pad as we have rarely seen this problem in the Qualtrics software used for testing.

Modifying the conversational language to be more direct eased the participant experience as they were able to understand who the question was asking about in each instance.

Some participants did overlook reading the italicized instruction at first but generally would return to reading the instruction later while answering. One participant mentioned that some people may overlook a ‘Do not include’ instruction that was included in one of the questions. Otherwise, the italicized instructions seemed to be informative to participants.

Participants did not comment on the number of questions per page but it did not seem to hinder their experience in completing the survey.

Participants were also told in retrospective probing that this survey is also administered via face to face/by telephone and were asked whether they would rather have completed this survey in one of those modes instead. Of the twenty participants asked, 16 said they preferred the web, three participants said face-to-face/phone, and one participant said either option was sufficient. Participants generally said they liked reading the questions and in certain circumstances, re-reading them if necessary:

- “I’m able to see and read questions and click on the answers.”
- “I wouldn’t want someone reading the questions to me and having them repeat themselves.”

Of the three participants who preferred face-to-face/telephone, two of them mentioned having an interviewer present to answer questions would be helpful to them:

- “If I have some doubts, you can answer the doubts.”

Overall, participants were able to answer the adapted self-administered NTIA Internet Use Survey successfully with few usability issues.

4.2 Types of Computing Devices Used

These questions cover the different types of devices a household uses, the performance of these devices, and the accessibility of these devices. There were eight questions administered in this section. We focus on six of the questions here that we asked retrospective probes about; there were two questions we did not probe on and therefore, do not present results here.

4.2.1 Introduction to Questions about Device Use (CMPINT)

Table 8. Introduction to Questions about Device Use

Final CAPI Production Wording	Self-Administered Tested Wording
This month we are asking some additional questions about Internet use. First, we will ask what types of computers and other devices [you/members of your household] currently use. Please focus on devices [you/members of your household] currently have access to, and have used at least occasionally during the past six months. We are interested in devices used at home, work, school, a library, or any other place.	This month we are asking some additional questions about Internet use. First, we will ask what types of computers and other devices members of [you/your household] currently use. [If multi-member household: You will be asked who in your household uses each device]. Please focus on devices [you/members of your household] currently have access to, and have used at least occasionally during the past six months. We are interested in devices used at home, work, school, a library, or any other place.

In CAPI testing, participants generally were thinking about the correct time frame even though we did not probe on this specifically. One participant did seem to answer based on what devices household members owned, which is incorrect as they should have been answering based on use.

We added the sentence “You will be asked who in your household...” to the intro for Internet because we wanted to explain to people in multi-member households what they would have to do for the following questions. In self-administered testing, one participant tended to omit devices that they did not necessarily own but still used. This participant mentioned watching a movie on their roommate’s Roku but they did not include themselves when answering TVBOX (this is also mentioned in the TVBOX findings). One participant in retrospective probing seemed unsure whether to answer these questions based on usage or actual ownership, but the participant correctly said they answered based on usage:

“Since question was asking based on who does utilize or use these things”.

Although a few participants were not clear that they only need to “use” a device rather than “own” a device to include it, this is not a finding specific to the Internet mode and we think this is still the most optimal wording. One suggestion is to possibly emphasize the word “use” to be all capitalized. However, research has shown that emphasizing words may not always be helpful (Horwitz & Nichols, 2022).

4.2.2 Desktops and Laptops (DESKTP and LAPTOP)

Table 9. Desktops and Laptops

Final CAPI Production Wording	Self-Administered Tested Wording
LAPTOP [Do you/Does anyone in this household, including you,] use a laptop computer at least occasionally? Include any laptop computers [you/members of your household] currently have access to and have used during the past six months. <i>(If needed)</i> A laptop is a personal computer that is portable and includes a built-in keyboard and screen. <i>(If yes & is multi-person household)</i> Who is that?	LAPTOP [Do you use/Who in your household uses] a laptop computer at least occasionally? Include any laptop computers [you/members of your household] currently have access to and have used during the past six months. • <i>A laptop computer is a personal computer that is portable and includes a built-in keyboard and screen.</i> • <i>Select all that apply.</i>
DESKTP What about a desktop? [Do you/Does anyone in this household] use a desktop computer? <i>(If needed)</i> A desktop is a personal computer that must be plugged into a power outlet and is typically at a single location. <i>(If yes & is multi-person household)</i> Who is that?	DESKTP [Do you use/Who uses] a desktop computer? • <i>A desktop computer is a personal computer that must be plugged into a power outlet and is typically at a single location.</i> • <i>Select all that apply.</i>

In CAPI testing, the desktop question preceded the laptop question for many of the interviews. A few participants incorrectly reported they had a desktop computer, so CBSM modified the question order for the laptop question to precede the desktop question. After this question order change, no participants incorrectly reported they had a desktop, even though we only conducted four interviews with this new order (Katz et al., 2024)

We further monitored this order change in self-administered testing. In self-administered testing, all participants except one seemed to answer this question accurately given their current situations. One participant mentioned they used the desktop once at a school library but did not want to use it anymore so it was not even occasional use for the person (they incorrectly answered “No” for this question).

All of the other participants generally were able to describe what a desktop computer was and most participants believed the question instruction text aligned with how they thought of a desktop. A couple participants also added that desktops often have a separate monitor, keyboard, and mouse.

4.2.3 Wearable Devices (WEARAB)

Table 10. Wearable Devices

Final CAPI Production Wording	Self-Administered Tested Wording
[Do you/Does anyone in this household] use a wearable device that is connected to the Internet, such as a smart watch or fitness band? <i>(If needed)</i> Examples include an Apple Watch or Fitbit. <i>(If yes & is multi-person household)</i> Who is that?	[Do you use/Who uses] a wearable device that is connected to the Internet, such as a smart watch or fitness band? • <i>Examples include an Apple Watch or Fitbit.</i> • <i>Select all that apply.</i>

As in CAPI testing, all participants were able to answer this question without much problem and only one participant in self-administered testing mentioned this question was difficult to answer. This participant could not remember if a roommate had a wearable device.

One participant mentioned that their small child has a doctor-recommended wearable device used for sleeping to monitor their heart rate. This device is usually tied to the leg but was sometimes tied around the child's hand. Another participant mentioned they had a Wo-Fit band so they reported 'Yes' but they speculated others may not report 'Yes' if they have a different type of fitness band that is not an Apple Watch/Fit Bit.

We did not recommend any changes to the self-administered wording for future testing.

4.2.4 Devices Played Through a TV (TVBOX)

Table 11. Device Played Through a TV

Final CAPI Production Wording	Self-Administered Tested Wording
[Do you/Does anyone in this household] use a smart TV, a game system, or another device that connects to the Internet and plays through a TV? Examples include an Apple TV, Xbox, or Roku that can access the Internet. <i>(If yes & is multi-person household) Who is that?</i>	[Do you use/Who uses] a smart TV, a game system, or another device that connects to the Internet and plays through a TV? • <i>Examples include an Apple TV, Xbox, or Roku that can access the Internet.</i> • <i>Select all that apply.</i>

Katz et al. (2024) found that a few participants in CAPI testing had trouble answering this question because they were unsure what constituted a smart TV/device that plays through a TV.

For self-administered testing, a few participants struggled with answering this question. One participant had a Nintendo Switch that was handheld but had never used it on the TV (they ultimately did not count this which was correct). A second participant was not familiar with a smart TV and mentioned that his roommate may have used a smart TV. Based on the description the participant provided, it was most likely a smart TV (this participant reported “No one in this household”).

A third participant lived with their dad who does not use the remote to control the smart TV. However, he still watches TV so the participant marked the dad as using a smart TV/device; this participant answered correctly. Also, another participant was wondering whether someone who is present watching TV but not actively “using” it is also in scope.

A fourth participant reported ‘No’ in the survey but mentioned in probing they would say ‘Yes’ because they connect their laptop to the television (which is not in scope). However, the participant later said that they had a “Xfinity Flexbox” but they preferred using the laptop through the TV. After the interview, we looked up the device. Xfinity Flex seems to be a 4K streaming TV box where you can stream movies/television shows and apps such as Netflix, Disney+, etc. Based on this, it seemed like the participant should have said ‘Yes’ but was not thinking about the “Xfinity Flex” while completing the survey.

During retrospective probing, the one participant who mentioned having a Switch said he and his roommate watched a movie on their other roommate’s Roku. However, he did not self-report that the two of them used a device to play through the TV because they personally didn’t own the device:

“I decided to answer in the way that was analogous to the other questions. Who owns one of these devices”.

This was the same participant who also mentioned using a desktop once but did not count that when answering the question.

The participant who was not familiar with the smart TV said they rarely watched TV and it is usually on their laptop. They said their roommate might have a Firestick but were not sure how often it was used or the utility of the device.

When asked about other devices that connect and play through TV, participants mentioned Wii/Nintendo Switch, Firesticks, and Chromecast. One participant mentioned DVDs which was out of scope.

If “passive use”, for example, watching a smart TV but not using a remote, is in scope for this question, this question may need to be reworded. It seems the idea of “using” versus “watching” TV could be confusing to participants. One possible rewording could be:

“Who watches or uses a smart TV, game system, or another device that connects to the Internet?”

4.2.5 Performance of Devices (DEVQUA)

Table 12. Performance of Devices

Final CAPI Production Wording	Self-Administered Tested Wording
Thinking about all the different devices we just discussed, how well do the computers and other Internet-connected devices [you use/used by this household] work overall? Please do not include home Wi-Fi routers or similar equipment. <i>Read and select best match; if respondent is unsure, ask for an average.</i> (1) They do not work at all (2) They work poorly (3) They somewhat work (4) They work well (5) They work perfectly	Thinking about all the different devices [you use/used by this household], how well do the computers and other Internet-connected devices work overall? • <i>Please do not include home Wi-Fi routers or similar equipment.</i> (1) They do not work at all (2) They work poorly (3) They somewhat work (4) They work well (5) They work perfectly

In CAPI testing, Katz et al. (2024) found that most participants were able to answer this question without much issue. However, a few participants included external factors like weather and Internet outages.

In self-administered testing, all participants either said “they work well” or “they work perfectly”. Many participants still seemed to be thinking about external factors that affect device usage. One participant was thinking about the “faulty” router they have. A second participant mentioned that sometimes the television lags because of speed, connection, or interference from a neighbor. A third participant mentioned thinking about weather affecting device usage that they can’t control and a fourth participant thought about infrequent disruptions like an outage affecting the devices.

One participant mentioned in retrospective probing being confused because of the word ‘Wi-Fi’ being in italicized text. The participant said, “everything is connected to Wi-Fi, so that threw me off”. The participant then asked:

“How well does it work in general or how well do they work when connected to the Wi-Fi?”

After re-reading the question, the participant then thought about the devices that were on Wi-Fi when answering this question (they reported “they work well”).

Several participants noted that while their devices function as intended nearly all of the time, they would never respond that the devices work perfectly. One said, “Perfectly is a pretty high standard and nothing is perfect, so they work well.”

Participants were able to define what “they work poorly” meant to them in this question. A few participants mentioned the actual device not working:

“[They] fail to function or that they crash or simply don’t have the capacity to do the job.”

Many participants in this round also discussed the speed or connection:

“I don’t get anything at all, no access at all. They don’t work when you need them to work. They don’t produce the amount of speed you need, it’s constantly streaming. That’s when I say it’s not functioning properly.”

We did not recommend any changes to the self-administered wording at this time, though the cross-mode issue of disentangling external events affecting device usage will need to be considered for future testing.

4.2.6 Inaccessibility of Devices (DEVSTA)

Table 13. Inaccessibility of Devices

Final CAPI Production and Self-Administered Tested Wording
At any point during the past six months, were [you/members of your household] unable to use a computer or other Internet-connected device because it was in use by someone else, located elsewhere, or otherwise inaccessible? (1) Yes (2) No

In CAPI testing, participants generally were able to answer this question without much issue (Katz et al 2024). A couple of participants did not provide codeable responses initially but eventually provided an answer that seemed to accurately reflect their situation.

In self-administered testing, seven participants reported ‘Yes’ and thirteen participants reported ‘No’. Most of the participants who reported ‘Yes’ verbalized their reasoning during the survey for answering the way they did, which all seemed to be in scope for this question. However, one participant mentioned that most of the time it is because the company (e.g., Verizon, Comcast) is having the issues; this was out of scope.

One participant mentioned that they take turns watching the TV, so they answered ‘Yes’ because they do not have a perfect schedule for everyone to access the TV (they have a smart TV). A second participant mentioned there were times they could not connect to the Internet completely. A third participant mentioned that there is one iPad for everyone to use and when someone is using it, it is inaccessible to everyone else, so they reported ‘Yes’. Another participant said during retrospective probing they thought about wind storms causing power outages and therefore not allowing the household to use the devices.

A few participants struggled answering the question. One participant said the following about “inaccessible”:

“What does inaccessible mean? Behind a locked door? What are we supposed to get from that? Unless it was actually left somewhere and we couldn't get it.”

The participant mentioned a story about one of her children locking a computer behind a door and not being able to get it. The participant ultimately decided to report ‘Yes’ which was the correct response in this situation. Similarly, during retrospective probing, one participant mentioned that the word “inaccessible” is a vague term. Another participant mentioned that “someone else” is a little confusing. Originally, they reported ‘No’:

“When I’m reading this question, I’m thinking that it’s saying did somebody else have my equipment and the answer is ‘No’.”

Later in retrospective probing, the participant was then thinking about an example of letting someone else use their phone to make a phone call so they should have said ‘Yes’ (it seems like the participant did answer the question correctly originally before they thought about this with ‘No’).

Even with the one participant thinking about a cable or internet provider when they were not supposed to, we still think this is the most optimal wording but this question should continue to be monitored in future testing.

4.3 Location of Internet Use

These questions cover the different locations where a household uses the Internet. There were eight questions administered in this section. We focus on two of the questions here that we asked retrospective probes about; there were six questions we did not probe on and therefore, do not present results here.

4.3.1 Introduction to Questions about Internet Location Use (INTINT)

Table 14. Introduction to Questions about Internet Location Use

Final CAPI Production Wording	Self-Administered Tested Wording
Next, we will ask where [you/members of your household] currently use the Internet. Please focus on places where [you/members of your household] have used the Internet at least occasionally during the past six months. Internet use includes a wide variety of activities on all devices, from checking email or browsing the Web to watching videos or using mobile apps.	Next, we will ask where [you/members of your household] currently use the Internet. (If multiple members in household: You will be asked who in your household uses the Internet at each location). Please focus on places where [you/members of your household] have used the Internet at least occasionally during the past six months. Internet use includes a wide variety of activities on all devices, from checking email or browsing the Web to watching videos or using mobile apps.

In CAPI testing, all participants understood the instruction as written (see Katz et al 2024). In self-administered testing, all participants understood this instruction as written including the slight changes made to transform the question from CAPI to a self-administered format. We added the sentence, “You will be asked who in your household...” to the intro for self-

administered because we wanted to remind participants what they would have to do for the following questions.

When asked which devices they were thinking of when answering this series of questions, most participants said they were thinking largely of cellphones, laptops, and tablets.

Some participants may have been thinking of only accessing the Internet through Wi-Fi and answered to that effect. For example, for the question that asked whether someone used the Internet at someone else’s home, one participant was only thinking of accessing Internet via Wi-Fi connection and was not sure whether their two housemates had access to Wi-Fi networks in the homes they visited. As a result, this participant chose to not report that their housemates were using the Internet in another person’s home which we think was the incorrect interpretation.

Since participants were wondering whether they needed to be connected to Wi-Fi for their internet use to count as being in that specific place, we recommend adding the following highlighted phrase to the instruction to help ease potential confusion for participants:

- “Next, we will ask where [you/members of your household] currently use the Internet. [If multi-member household: You will be asked who in your household uses the Internet at each location.] Please focus on the places where [you/members of your household] have used the Internet at least occasionally during the past six months. **Internet use can be on Wi-Fi or through cellular data** and includes a wide variety of activities on all devices, from checking email or browsing the Web to watching videos using mobile apps.

NTIA also suggested for future testing to put in an instruction after the question to ‘include all smartphones’. Both of these recommendations will be evaluated in future testing.

4.3.2 Using Internet From Place to Place (INTRAV)

Table 15. Using Internet From Place to Place

Final CAPI Production Wording	Self-Administered Tested Wording
What about while going from place to place? Examples include using the Internet while riding public transit, or using Google Maps or Waze on your cell phone while driving. <i>(If needed)</i> [Do you/Does anyone in this household] use the Internet while going from one place to another? <i>(If yes & is multi-person household)</i> Who is that?	Who uses the Internet while going from place to place? • <i>Examples include using the Internet while riding public transit, or using Google Maps or Waze on your cell phone while driving.</i> • <i>Select all that apply.</i>

As in CAPI testing, participants did not struggle with this question generally. In probing, participants were asked to rate the difficulty they experienced with this question. Most felt the question was relatively easy to answer. A majority of participants noted that they use their phones while on public transit or driving around for maps, music, or communicating with others.

One did ask if they were driving, and their spouse was in the car using a map app to navigate, if they would both be using the Internet in transit. They ultimately decided that they were both using the Internet in this situation and correctly chose 'Yes' in response.

We did not recommend any changes to the self-administered wording for future testing.

4.4 Type of Internet Service Used

These questions cover the different locations where a household uses the Internet. There were eight questions administered in this section. We focus on one of the questions here that we asked retrospective probes about; there were seven questions we did not probe on and therefore, do not present results here.

4.4.1 Home Internet Service (HOMTE)

Table 16. Home Internet Service

CAPI Production Wording	Self-Administered Tested Wording
I am going to read a list of ways that people access the Internet from their homes, other than a mobile data plan. At home, [do you/does anyone in this household] access the Internet using: <i>Read and select all that apply.</i> (1) High-speed Internet service installed at home, such as cable, DSL, or fiber-optic service? (If needed) This type of Internet service is often provided by a cable company or phone company. (2) Satellite Internet service? (If needed) This type of Internet service is received through a satellite dish installed at home. (3) Some other service? <i>(Do not read) None</i>	The following questions ask the different ways people access the Internet from their homes, other than a mobile data plan. At home, [do you/does anyone in this household] access the Internet using: 1) High-speed Internet service installed at home, such as cable, DSL, or fiber-optic service? • <i>This type of Internet service is often provided by a cable company or phone company.</i> (1) Yes (2) No 2) Satellite Internet service? • <i>This type of Internet service is received through a satellite dish installed at home.</i> (1) Yes (2) No 3) Some other service? (1) Yes (2) No

In CAPI testing, participants struggled answering the question that had been used in prior production cycles. We also tested a modified version of the response option “High-speed Internet service...” in the first round but it did not perform any better. Therefore, we kept the CAPI production wording that had been used in the past.

It was easier for participants to answer this question in a self-administered format; only a couple of participants struggled answering this question. All participants reported they have “High-speed Internet service...”, one participant reported they have “Satellite Internet service”, and one participant reported they had “Some other service”. The one participant who reported “Satellite Internet Service” said they had Satellite TV but were not sure what company it was so it is unclear whether they actually had Satellite Internet service. The participant who said “Some other service” mentioned having a “box” that Xfinity had put in the home (this seems like it was not a separate Internet service provided).

When answering this question, most participants were able to easily report they had High-Speed Internet service. The examples in the “High-speed Internet service...” option generally assisted participants in answering. One participant in retrospective probing knew they had

High-speed Internet service that was installed by the landlord but they were not sure whether it was cable, DSL, or fiber optic. Another participant had broadband and assumed that is what we were asking about but mentioned someone else may mark 'No' if they had broadband (this participant seemed to answer correctly).

A few participants were confused about "some other service". Two participants said they wished there were examples (one participant was not able to provide any examples when asked). Another participant said:

"Not sure what some other service means. Besides cable company what else is there?"

When participants were probed about what constituted "some other service", one participant said "anything other than cable or dish". A couple of participants mentioned hot spots. A couple of participants did not provide in-scope answers, with one participant mentioning using a phone carrier like T-Mobile/Verizon and another participant mentioning just other locations they can use the Internet.

Table 17. Alternate Home Internet Service Wording

Self-Administered Tested Wording	Alternate wording shown during retrospective probing
The following questions ask the different ways people access the Internet from their homes, other than a mobile data plan. At home, [do you/does anyone in this household] access the Internet using: 1) High-speed Internet service installed at home, such as cable, DSL, or fiber-optic service? • <i>This type of Internet service is often provided by a cable company or phone company.</i> (1) Yes (2) No 2) Satellite Internet service? • <i>This type of Internet service is received through a satellite dish installed at home.</i> (1) Yes (2) No 3) Some other service? (1) Yes (2) No	Who is your home Internet Service Provider? (1) Xfinity (2) Spectrum (3) AT&T (4) Verizon (5) Cox (6) Altice (7) CenturyLink or Quantum Fiber (8) Frontier (9) T-Mobile (10) Other-please specify: _____ How does AT&T provide the Internet service you use at home? • <i>Select all that apply.</i> (1) A wire runs into my home and is connected to a modem or router. (2) My/our mobile phone(s) or other devices connect directly to AT&T's wireless network (not via Wi-Fi), just like outside of home How does Verizon provide the Internet service you use at home? • <i>Select all that apply.</i> (1) A wire runs into my home and is connected to a modem or router. (2) My/our mobile phone(s) or other devices connect directly to Verizon's wireless network (not via Wi-Fi), just like outside of home. (3) Verizon provided me/us with a device that connects to their wireless network and creates a Wi-Fi network for my/our home. How does T-Mobile provide the Internet service you use at home? • <i>Select all that apply.</i> (1) My/our mobile phone(s) or other devices connect directly to T-Mobile's wireless network (not via Wi-Fi), just like outside of home (2) T-Mobile provided me/us with a device that connects to their wireless network and creates a Wi-Fi network for my/our home.

After probing about the current Home Internet Service question, participants were administered another question about the type of Internet Service provider they had and follow-up questions about how their Internet service was provided if they had reported they had AT&T, Verizon or T-Mobile. These questions were conducted via screensharing- the participant could see the question and the interviewer selected the answers the participant reported aloud.

Of the nineteen participants who answered this question, eight participants reported they had Xfinity, six participants reported they had Verizon, two participants reported they had Spectrum, two participants reported they had another Internet Service Provider, and one participant did not know what other service they had (they marked "Other" and wrote "I don't know").

Among the six participants who reported they had Verizon, three participants reported "A wire runs into my home..", one participant reported "My/our mobile phone(s) or other devices connect directly to Verizon's wireless network...", and two participants reported "Verizon provided me/us with a device that connects to their wireless network and creates a Wi-Fi network for my/our home."

One participant noticed T-Mobile on there and wanted to choose that answer along with Verizon since T-Mobile was his phone carrier. When probed because he could only select one choice, the participant chose Verizon since that is the home Internet Service Provider which seemed to be the correct answer. Two participants said the "Verizon" option should have "Fios" added in while one participant recommended rewording "Xfinity" to "Comcast Xfinity". One participant who reported "A wire runs into my home..." in the Verizon follow-up question reported having a wireless router, so this seemed like the closest option. They added that someone came to the home to set it up and plug the wire into the wall.

Another participant who reported "A wire runs into my home..." indicated that the router they have is connected to a modem provided by Verizon. The participant found the third option, "Verizon provided me/us with a device...", confusing. The device was owned by Verizon so it seems like the participant did not understand the response option fully.

One participant who reported "Verizon provided me/us with a device..." mentioned Verizon gave the household a router that plugged into the wall. They added that the wire for the modem goes into a battery (not connected to the wall) but then the router plugs into an electric wall socket.

Of the thirteen participants that were asked whether they preferred the original home Internet service question or the Internet service provider question, six reported home Internet service, seven reported Internet service provider, and one did not have a preference.

Given that HOMTE performed much better in a self-administered setting, we recommend using that wording for future self-administered testing. Although this question has been found to be difficult in CAPI for a few participants, it is encouraging that so far self-administered participants have been able to answer this. Future testing will need to continue to evaluate this question and alternatives in CAPI. The Pew Research Center is also researching in how to measure home broadband use and subscription in self-administered surveys (McClain et al., 2024).

We also found that it may be tricky to administer questions about the actual Internet service provider in general. Although participants were able to identify their provider, it may be difficult for them to describe what type of Internet service they have. NTIA will need to decide what level of detail they want because that could affect comprehension of the question.

4.5 Computing Applications and Services Used

These questions cover the different types of Internet applications and services used. There were 19 questions administered to the participant either for themselves or as a proxy answering about the randomly selected household member; then four questions were administered about the household’s use of medical applications and services. We focus on five of the questions here, since there was either retrospective probes administered following completion of the survey or interest in how participants interpreted these questions. There were 18 questions we did not probe on and therefore, do not present results here.

4.5.1 Video or Voice Calls (CONFER)

Table 18. Video or Voice Calls

CAPI Production Wording	Self-Administered Wording
In the past six months, [have you/has NAME] participated in video or voice calls or conferencing over the Internet, such as with FaceTime or Zoom? <i>(If needed)</i> [Do you/Does NAME] participate in video or voice calls or conferencing? (1) Yes (2) No	[Do you/Does NAME] participate in video or voice calls or conferencing over the Internet, such as with FaceTime, Zoom, or WhatsApp? (1) Yes (2) No

Overall, participants found this question easy to answer. During CAPI testing, participants described CONFER as providing “helpful examples” of video conferencing platforms like Facetime and Zoom. During retrospective probing, participants were asked if there were any video call applications they were thinking about that were not mentioned in the question wording. Microsoft Teams and Google Meet were common voice call apps mentioned. Several mentioned using WhatsApp in both CAPI and self-administered testing, where one participant said they used the platform specifically for video/voice calls with their non-American friends.

In CAPI testing, one participant suggested we consider asking if people use Zoom and Teams for work meetings and video calls and other platforms like Facetime and Facebook video for personal use. In that case, using conferencing apps for work versus personal use seemed like different concepts for them. In the self-administered testing, one participant had a similar comment and suggested we include Microsoft Teams for work as one of the examples. They said, “People may say 'No' because they may not think to consider Teams in their answer if you don't put Teams.” We do not think this is a significant issue, because two participants who said they only use Microsoft Teams for work but not any of the other examples still answered ‘Yes’ to the question without difficulty.

When asked if there were any other platforms people thought about that were not included in the question in retrospective probing, participants mentioned the Discord app on the phone and PC, Google Meet, Slack, Webex, and telehealth specific conferencing for virtual appointments with their doctor. One participant said they “rarely” do video conferencing and it “happens only once or twice in a blue moon” so they answered ‘No’.

We recommended putting the examples on a separate line and in bullet format for consistency with other questions. Therefore, it would read:

[Do you/Does NAME] participate in video or voice calls or conferencing over the Internet?

- *Examples include FaceTime, Zoom, or WhatsApp.*

(1) Yes

(2) No

NTIA accepted this recommendation to test in future self-administered testing.

4.5.2 Streaming or Downloading Videos (VIDEO)

Table 19. Streaming or Downloading Videos

CAPI Production Wording	Self-Administered Wording
What about streaming or downloading videos? [Do you/Does NAME] watch videos over the Internet? Examples include YouTube and Netflix. (1) Yes (2) No	[Do you/Does NAME] watch videos over the Internet? • <i>Examples include streaming or downloading videos on YouTube and Netflix.</i> (1) Yes (2) No

All participants were able to answer this question; however, during retrospective probing some hesitated when asked about whether using the word “streaming” or “downloading” versus “watching” videos would have made the question easier to answer. One participant said, “The word download trips me up a little bit because I definitely don’t download videos off YouTube. I maybe have done it from Netflix twice in my life when I was about to get on a plane.”

Similarly, another participant said, “Watching videos over Internet is easier to understand. Most people use streaming for Netflix...streaming might also be watching something live like football which has a different connotation than watching.” When asked about the phrase ‘Watch videos’, two participants thought of watching videos on TV, YouTube and TikTok, which they considered separate from streaming movies and downloading shows. When asked about the word ‘streaming’, one person said, ‘Netflix shows aren’t videos’, but was still able to answer ‘Yes’ to the question without difficulty. Generally, some participants thought this question was “too broad”.

Despite these comments, most participants said they liked the current wording of VIDEO, and that streaming/downloading mean the same thing as watching/listening so they were “interchangeable” to them. One participant mentioned that “Some people could watch videos that they took on their phone and stream Netflix on the phone app. Same thing”. Another commented, “Streaming and watching seem like the same things. Streaming is an example here, but it was clear that the question is all encompassing of what streaming videos and watching videos mean.” Several participants said that “streaming and downloading” were more

specific and modern, while “watching videos” is a more general term, however both make sense and meant the same thing.

Although some participants may be confused about whether there are differences between watching and streaming/downloading, we still think this is the most optimal wording as participants are still able to answer this question without much issue. We recommend monitoring this wording in future testing to ensure that including both watching and streaming/downloading does not present any difficulties; NTIA agreed with this recommendation.

4.5.3 Telework (TELEWK)

Table 20. Telework

CAPI Production Wording and Self-Administered Wording	Wording Recommended for Future Testing
[Do you/Does NAME] use the Internet to telecommute or work while away from [your/his/her] usual workplace, such as working from home? (1) Yes (2) No	[Do you/Does NAME] use the Internet to telecommute or work while away from [your/his/her] usual workplace, such as working from home? (1) Yes (2) No

Generally, there was no difficulty with the question. Similarly to CAPI testing, participants considered the nature of their household member’s job before answering the question. One participant paused for a few seconds and thought about whether they use the Internet as a warehouse worker. This participant answered ‘No.’

Several participants seemed to think back to their answers for INWORK ([Who uses]/[Do you use] the Internet at work?) when answering this question. One participant answered, ‘Yes’ to TELEWK and mentioned that their spouse works from home 100% remotely, but they personally have a hybrid work structure. This person said “the wording on this [TELEWK] feels exclusionary for people who only work remotely. I’m still going to answer ‘Yes’ to this”, which seemed to be the correct answer. This person thought

the question seemed to exclude people who work 100% from home because home is their ‘usual workplace’ rather than a temporary workplace as the question suggests.

To help participants comprehend this question, we modified the question to be the following for future testing:

“[Do you/Does NAME] use the Internet to telecommute or work from home?”

4.5.4 Financial Services (FINANC)

Table 21. Financial Services

CAPI Production Wording	Self-Administered Wording
[Do you/Does NAME] use the Internet for financial services such as online banking, investing, paying bills, or sending money to other people using services like CashApp, Venmo, or PayPal? <i>(If needed)</i> Examples include using your bank’s website or app, and paying a credit card or utility bill online. (1) Yes (2) No	[Have you/Has NAME] used the Internet for financial services such as online banking, investing, paying bills, or sending money to other people using services like CashApp, Venmo, or PayPal? • <i>Examples include using your bank’s website or app, and paying a credit card or utility bill online.</i> (1) Yes (2) No

Most participants did not have any difficulty answering this question and mentioned various services used during the think-aloud, including paying rent to their landlord online and online banking. In both CAPI and self-administered testing, all participants mentioned at least one of the peer-to-peer transaction apps in our examples, such as CashApp or Venmo. Zelle was also another application mentioned where individuals send each other money.

In self-administered testing, several participants answered this question with a slightly negative tone and shared that they reluctantly use financial services. One person who labeled their spouse as ‘digital resistant’ said, “Not so much investing, she hates paying bills online and we don’t do any of those apps like Venmo. But boom yes, we do go to the bank’s website.” This person correctly answered ‘Yes’ to the question. The participant noted that both the stem and bullet points have examples and thought it read weird. They suggested that we put them in one place or another. We had moved the money distribution services examples from the instruction

to question stem since one participant in CAPI testing erroneously omitted that when answering.

Another person who answered ‘Yes’ said ‘Everyone is forced to do financial things like online banking nowadays’. These responses seemed to reflect participant’s views on technology and Internet use in general, rather than issues with the question.

Although one participant did mention having examples in both the question text and instructions was strange, we recommended keeping the examples in the question text for future testing given what we found in CAPI testing, to which NTIA agreed.

4.5.5 Household Equipment or Appliances (HOMIOT)

Table 22. Household Equipment or Appliances

CAPI Production Wording	Self-Administered Wording
What about interacting with household equipment or appliances that are connected to the Internet, such as a connected thermostat, light bulb, or security system? <i>(If needed)</i> [Do you/Does NAME] use the Internet to interact with household equipment or appliances?	[Do you/Does NAME] interact with household equipment or appliances that are connected to the Internet, such as a smart thermostat, light bulb, or security system? • <i>Examples include nest thermostat and ring camera.</i> (1) Yes (2) No

Prior to self-administered testing, we added the examples “Nest Thermostat” and “Ring camera” to assist respondents in answering this question.

Most participants appeared to answer this question correctly; however, similarly to CAPI testing, several participants seemed to have some difficulty in interpreting the question. One participant who answered ‘No’ has a Google Home, but does not use it to operate another device, such as to turn on any appliances like the lights or a thermostat. They described it as a standalone device that can stream music or connect through Bluetooth. This participant seemed to answer correctly.

Another person said the list of items in the examples ‘made it easy to answer’ and they answered ‘No’ for this question. This is because they would rather use the original on/off switch

for their home devices, instead of using the app connected to their phone that can control them.

All were familiar with the Ring and Nest examples and most participants mentioned using Alexa to control the smart plug that connects to the lights and other appliances. One participant suggested that the term 'smart plug' should be added to the list of examples here. They said, "Devices may not be considered 'smart'. It's a regular lamp but the smart plug allows me to interact it with and control it via my voice/phone."

Overall, in self-administered testing, the majority of those who thought the question was easy to answer all had smart devices that they used often, while a few thought the question was easy because they have never had any smart devices in their home.

Even with several participants struggling, we still think the tested question wording is the most optimal for the self-administered mode. The examples added seemed to help participants with answering. We recommended probing on "smart plug" in the future to see if participants are familiar with that term.

4.6 Challenges and Concerns With Internet Use

These questions cover the different types of services about the challenges and concerns the household has about using the Internet. There were seven questions administered in this section at the household level. We focus on two of the questions here that we asked retrospective probes about; there were five questions we did not probe on and therefore, do not present results here.

4.6.1 Concerns about Privacy and Security (PSPRE)

Table 23. Concerns about Privacy and Security

CAPI Production Wording	Self-Administered Tested Wording
During the past 12 months, have concerns about privacy or security stopped [you/anyone in this household] from: <i>Read and select all that apply</i> (1) Conducting financial transactions such as banking, investing, or paying bills online? (2) Buying goods or services online? (3) Posting photos or other information on social networks? (4) Expressing an opinion on a controversial or political issue, such as on a blog or social network, or in an email? (5) Finding information using Google or another search engine? (Do not read) None of the above	During the past 12 months, have concerns about privacy or security stopped [you/anyone in this household] from: a. Conducting financial transactions such as banking, investing, or paying bills online? (1) Yes (2) No b. Buying goods or services online? (1) Yes (2) No c. Posting photos or other information on social networks? (1) Yes (2) No d. Expressing an opinion on a controversial or political issue, such as on a blog or social network, or in an email? (1) Yes (2) No e. Finding information using Google or another search engine? (1) Yes (2) No

In self-administered testing, we modified this item to a Yes/No question-by-question format.

In CAPI testing, there were a few problems participants had with answering the question. This included participants needing the question re-read, not thinking of the correct time frame, and not answering the question in terms of privacy or security concerns (Katz et al., 2024).

All participants were able to answer this question in self-administered testing, but similar issues found in CAPI testing emerged. A couple of participants had to scroll back up to re-read the main question wording when answering the sub-questions. Additionally, a couple of participants initially had trouble comprehending the question. One participant said the following when answering (1):

“Were you asking about whether there is the concern about distrust of using services online or a distrust of using the financial services over a shared Internet connection with roommates? We all still do it anyway, regardless of what it’s asking. So I’m answering ‘No’.”

This participant seemed to answer the question correctly.

A second participant said that we may need to emphasize “stopped” in the question. Initially, the participant read it as “do we do these things: yes/no” but then had to go back and answered all of the sub-questions ‘No’. They said the following:

“Yes, we care about privacy, but no it hasn’t stopped us from doing any of these things”.

When participants were asked whether the question was easy or difficult during retrospective probing, one participant was not sure what was in scope for this question:

“[I] was unsure what counted as privacy/security concerns because that could be someone from social media is going to stalk you or it can be credit card fraud”.

Another participant seemed to misinterpret the question as they said when answering (3) they had good upload/download speeds. During retrospective probing, the participant mentioned that they were thinking more about the reliability of these activities and how well they worked rather than the privacy/security.

When asked about the timeframe, the majority of the participants said they were thinking about either the past 6 or 12 months. Other participants mentioned thinking about “any time” or the past three months.

Although some participants struggled with answering this question, there did not appear to be any new issues that came up when adapting this wording to self-administered. One suggestion is to possibly emphasize the word “stopped” to be all capitalized. However, research has shown

that emphasizing words may not be helpful in all cases. This will be re-evaluated in future self-administered testing.

A couple participants had to scroll up on the screen to re-read the question. Although this did not result in any participant errors, it is not ideal to have users scroll back and forth. One design change to explore in future testing is to adapt this question into a yes/no grid so the question wording and sub-questions display closer together to reduce the amount of participant scrolling. Another design alternative is to adapt this question to be ‘select all that apply’ and the sub-question wordings are modified to response options. Note, prior research has found that less response options are affirmatively reported in a ‘select all that apply’ design compared to a Yes/No layout.

4.6.2 Most Concerned Pertaining to Online Privacy and Security Risks (PSCON)

Table 24. Most Concerned Pertaining to Online Privacy and Security Risks

CAPI Production Wording	Self-Administered Tested Wording
Overall, what concerns [you/members of this household] the most when it comes to online privacy and security risks? <i>Do not read; select all that apply and/or enter verbatim response if other</i> (1) Identity theft (2) Credit card or banking fraud (3) Data collection or tracking by online services (4) Data collection or tracking by government (5) Loss of control over personal data such as email or social network profiles (6) Threats to personal safety, such as online harassment, stalking, or cyber-bullying (7) No concerns (8) Other: _____	Overall, what concerns [you/members of this household] the most when it comes to online privacy and security risks? • <i>Select all that apply</i> (1) Identity theft (2) Credit card or banking fraud (3) Data collection or tracking by online services (4) Data collection or tracking by government (5) Loss of control over personal data such as email or social network profiles (6) Threats to personal safety, such as online harassment, stalking, or cyber-bullying (7) No concerns (8) Other – please specify: _____

The CAPI production wording is currently an open-ended question for which interviewers code verbatim responses to a list of predefined categories as shown to the left. Participants generally did not have issues answering this question in CAPI. Below was the breakdown of answers:

- 1) Identity theft-10
- 2) Credit card or banking fraud-4
- 3) Data collection or tracking by online services-5
- 4) Data collection or tracking my government-2

- 5) Loss of control over personal data such as email or social network profiles-3
- 6) Threats to personal safety, such as online harassment, stalking, or cyber-bullying-0
- 7) No concerns-1
- 8) Other-7

For self-administered, we modified this question to be a “Select all that apply”. In self-administered testing, participants did not have issues answering this question. One participant had trouble selecting the options but this happened in other questions in the survey so it may have been something with the participant’s mouse/touch pad. Participants generally seemed to select more response options which makes sense considering they were able to see all of the options. Below was the breakdown of answers:

- 1) Identity theft-13
- 2) Credit card or banking fraud-14
- 3) Data collection or tracking by online services-14
- 4) Data collection or tracking by government-8
- 5) Loss of control over personal data such as email or social network profiles-14
- 6) Threats to personal safety, such as online harassment, stalking, or cyber-bullying-5
- 7) No concerns-0
- 8) Other-1

The one participant who reported ‘Other’ typed in “phishing scams” and recommended to add more specificity to the category on threats. Another participant mentioned they were not concerned about government control until they saw the response option listed (they selected “Data collection or tracking by government”).

In retrospective probing, participants were asked whether the question was easy or difficult to answer. The majority of participants said the question was easy to answer. For the couple of participants that said the question was difficult, one said they had to think hard for the other people in the household so they thought about the conversations they have had in the past about privacy concerns. Another participant mentioned something similar, where they were not sure what their roommate does other than play games and watch YouTube (this person said the question was easy).

One participant said the question was easy but found it frustrating that they clicked ‘No concerns’ and it removed all of the other selected choices. Note, for future testing, this type of usability issue can be resolved by either (1) if a user would select one of the first seven options, it would gray out and disable ‘No concerns’ so no user can select that option or (2) adding an edit message to convey to the user to uncheck either ‘No concerns’ or the other selected options.

When asked if any of the categories were difficult, most participants said they understood what the categories meant. One participant mentioned that “data collection” was really the only one they had to focus on but they were not confused by it. A few participants seemed confused over the “loss of control over personal data...” option. One participant said:

- “Are you talking about when someone hijacks your account”?

A second participant said:

- “Would that mean like getting hacked on social media or again having your email address compromised?”

Another participant suggested rewording “loss of control over personal data...” to something like “hacking into profile for loss of control”.

Given that this was the first testing where these categories were read/displayed to the participant, the wording of these categories will need to be monitored in future testing. It will also be important to compare distributions of responses to this question in CAPI and Internet.

5. CONCLUSIONS

The goal of this exploratory testing was to explore the feasibility of a self-administered NTIA Internet Use Survey. Exploratory testing showed that global changes that were incorporated across the self-administered instrument were successful, including adapting the conversational language for self-administered and moving ‘if needed’ CAPI instructions to be on its own separate line in self-administered.

Most of the individual survey questions tested well in self-administered mode. Select questions that were found to be problematic in CAPI testing were also problematic in self-administered testing. This included questions asking about households using a smart TV, using household devices that are connected to the Internet like a smart thermostat or security system, and concerns using the Internet.

However, the question that asked about home Internet service was found to be less problematic for participants answering the survey in a self-administered mode compared to the CAPI mode. The home Internet service question may be complex to answer in a CAPI mode due to the extensive length; however, with the self-administered mode, it may be easier for the participant to comprehend by reading the question on screen. The question asking what concerns households the most in terms of online privacy and security risks had been an open-ended coded question in CAPI and was adapted to a mark all that apply in self-administered.

Participants understood this question in the new mode, but this question will still need to be monitored in the future to evaluate the data quality, given the change in question type.

Although the changes from CAPI to self-administered performed well in the exploratory testing, this is just the first step in exploring an Internet mode of the survey. For this study, we did not test a mobile version of the questionnaire as the goal was mainly to see how these questions would perform in a self-administered mode on a laptop/desktop computer. However, it is essential that surveys are now designed to be completed by mobile users; 32% of responding households completed the American Community Survey (ACS) on a mobile device in 2023 (Mills, 2024). Future research should explore cognitive and usability testing of adapted self-administered changes on mobile devices. In addition, field testing should be conducted to evaluate any potential mode differences between self-administered and CAPI.

Overall, other surveys should consider starting with similar exploratory cognitive and usability testing if they are thinking of adding a self-administered web component. This is a successful first step in understanding any possible differences in comprehension and reporting when adapting question wording across modes.

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Appendix A. NTIA Internet Use Survey Cognitive and Usability Interview Protocol

NTIA Internet Use Survey Cognitive Interview Protocol

Participant ID #: _____

Interview Date: ____/____/____ (mm/dd/yyyy)

Interviewer initials: _____

Start Time: _____ AM / PM End Time: _____ AM / PM

Double-check that you have the following information ready prior to the start of the interview:

- Household Roster
- List of relations of household members (up to 3) to the participant

MAKE SURE TO CONDUCT A “TECH CHECK” WITH PARTICIPANT FIVE MINUTES PRIOR TO INTERVIEW

SECTION I: INTRODUCTION

Hello, my name is _____, and I work for the Census Bureau. Thank you for agreeing to participate in our study. [IF OBSERVER PRESENT: I also want to let you know that this interview is being observed by [DESCRIBE OBSERVER’S ROLE: e.g., someone who works on the survey]. They are observing the interview because they are also interested in how the questions work for people like you.]

The Census Bureau counts the population in the U.S and also conducts different kinds of surveys.

Today, we are working on improving questions for a part of the Current Population Survey. You will help us test questions that ask about how members of your household use technology. It is important that these questions make sense, are easy to answer, and that everyone understands the questions the same way. Remember, there are no wrong answers, because only you know what you are thinking. Please answer these questions as you would if an interviewer had come to your home to complete the survey. Do you have any questions before we begin?

SECTION II: CONSENT FORM

- IF THEY DID SIGN BEFORE THE INTERVIEW, CHECK IN QUALTRICS

Before we start, I just wanted to thank you for signing the document in the link I sent before the interview. Just as a quick reminder, that document explained a little bit about this interview and

provided information about your rights as a participant, such as that all information you provide is confidential and we won't use your name or specific circumstances in any report. The link had a signature line to acknowledge the \$40 you will receive by mail for participating. It also asked for your permission to have this session audio recorded as well as a video recording of your computer screen and image. This is so that I can concentrate on what you are saying rather than having to take extensive notes. Your name will not be associated with the recording or any of the other data collected during the session. May I turn on the recorder? **[IF THEY SAY YES TURN ON SNAGIT RECORDER AND BACKUP AUDIO RECORDER]** For our purposes, we need to capture your consent orally on the recording. Please state, "I give consent to participate in this research and to be recorded."

- **IF THEY DID NOT SIGN BEFORE THE INTERVIEW:**

Before we start, I noticed that you had not yet signed the document in the link we e-mailed to you; I just wanted to check did you receive that e-mail? **[IF YES]** Did you have any issues opening the link or with signing the document? **[IF THEY SAY YES TALK THEM THROUGH IT. IF UNABLE TO GET SIGNED CONSENT, DISPLAY CONSENT FORM OR RELY ON ORAL CONSENT USING SCRIPT BELOW]** I will go over what is in that document and obtain consent orally. However, the document also contains an acknowledgement that you will receive \$40 by mail for participating in this research. You can still participate right now, but you will need to sign that document before we can send the \$40.

The document explains a little bit about the purpose of this study, which is to test questions for a part of the Current Population Survey. The document provides information about your rights as a participant, such as that all information you provide is confidential and we won't use your name or specific circumstances in any report. It also asks for your permission to have this session audio recorded as well as a video of your computer screen and image so that I can concentrate on what you are saying rather than having to take extensive notes. May I turn on the recorder? **[IF THEY SAY YES TURN ON SNAGIT RECORDER AND BACKUP AUDIO RECORDER]** For our purposes, we need to capture your consent orally on the recording. Please state, "I give consent to participate in this research and to be recorded."

- **PARTICIPANT AGREES**

Thank you for that.

Your participation in this interview is very important because it will help the Census Bureau with improving these survey questions. Your participation in this interview is voluntary, and you may decline to answer any question, or stop the interview at any time. We are scheduled to be here for about 65 minutes, and I won't keep you any longer.

- **IF PARTICIPANT IS NOT A FEDERAL EMPLOYEE:** As a thank you for your help, we will mail you \$40 cash after the interview is completed.

Do you have any other questions about the document or the interview?

SECTION III: INSTRUCTIONS AND SET UP

While you are answering questions for the survey, I would like you to think aloud. While I am interested in your answers, I am even more interested in the process you go through when you answer the questions. So, as much as possible, I'd like you to think aloud for me; meaning, talk me through your thought process as you answer each question. From time to time, I may also ask you what you are thinking to remind you to do this.

I find it helps people to get the idea of thinking aloud while they are answering a question if they do a practice question first. Let's begin with a practice question. Remember to think aloud as you answer.

How many windows are there in the house or apartment where you live?

_____ WINDOWS

IF NEEDED: *Try to visualize the place where you live, and think about how many windows there are in that place. As you count up the windows, tell me what you are seeing and thinking about.*

PROBES:

- *How did you come up with that answer?*
- *Tell me more about that. Why did you say [ANSWER]?*
- *I noticed that you hesitated. Tell me what you were thinking.]*

Great, that is what we will want you to do for today's session.

Do you have any other questions about the interview?

FOR WEB ONLY

- **ASK PARTICIPANT TO SHARE THEIR SCREEN.**
- **THEY NEED TO GIVE TEAMS PERMISSION TO SHARE THEIR SCREEN. IF THEY ARE ON A WORK COMPUTER, THEY MAY NOT BE ABLE TO SHARE THEIR SCREEN. THIS HAS TO BE DONE IN THEIR COMPUTER OR BROWSER'S SETTINGS. USUALLY TEAMS WILL PROMPT THEM TO DO THIS.**

Now I am going to send you a link to the survey and have you share your screen with me. Remember to close anything you don't want me to see, or you can only share your browser window

SECTION IV: HOUSEHOLD ROSTER

Ask R1 – R11 for 1st HH member. **Record information on Household Roster.** Then ask for next HH member. Repeat for every member of HH.

R1. NAME **What are the names of all persons living or staying here, starting with yourself?**

R2. SEX **What is [your/[NAME]'s] sex?**

1. Male
2. Female

R3. REL

IF NOT RESPONDENT: **How is [NAME] related to you?**

1. Opposite sex spouse (husband/wife)
2. Opposite sex unmarried partner
3. Same sex spouse (husband/wife)
4. Same sex unmarried partner
5. Child
6. Grandchild
7. Parent
8. Brother/sister
9. Other relative (aunt, uncle, cousin, in-law)
10. Foster child
11. Housemate/roommate
12. Roomer/boarder
13. Other nonrelative

R4. AGE **What is [your/NAME's] date of birth?**

R5. EDUCATION

What is the highest level of school [you have /[NAME has]] completed or the highest degree [you have/[NAME has]] received?

1. 1st, 2nd, 3rd, or 4th grade
2. 5th or 6th grade
3. 7th or 8th grade
4. 9th grade
5. 10th grade or 11th grade
6. 12th grade, no diploma
7. High school graduate-high school diploma or equivalent
8. Some college – no diploma
9. Associates degree – occupational / vocational program
10. Associates degree – academic program
11. Bachelor's degree (BA, BS, AB)
12. Master's degree (MS, MA, MBA, MEd, MEng, MSW)
13. Professional degree (MD, DDS, DMV, LLB, JD)
14. Doctorate degree (PhD, EdD)

R6. HISPANIC

(Are/Is)(Name/You) of Hispanic, Latino, or Spanish origin?

1. No, not of Hispanic, Latino, or Spanish origin
2. Yes, Mexican, Mexican Am., Chicano
3. Yes, Puerto Rican
4. Yes, Cuban
5. Yes, another Hispanic, Latino, or Spanish origin – *Enter, for example, Salvadoran, Dominican, Colombian, Guatemalan, Spaniard, Ecuadorian, etc.*

R7. RACEA

What is [NAME/your] race?

- 1. White**
- 2. Black or African American**
- 3. American Indian or Alaska Native**

4. Asian

5. Native Hawaiian or Other Pacific Islander

R8. WORK **LAST WEEK, did (name/you) do ANY work for either pay or profit?**

1. Yes
2. No

SECTION V: NTIA Internet Use Survey

FOR USEINT, ADMINISTER THE QUESTIONS ABOUT THE FOLLOWING

HH Member #1: **Name:** _____, **Relation to Participant:** _____,
HH Member #2*: **Name:** _____, **Relation to Participant:** _____,
HH Member #3*: **Name:** _____, **Relation to Participant:** _____,

[FOR QUESTIONS USEINT TO HOMIOT, SELECT HOUSEHOLD MEMBER THAT WAS ASSIGNED BY PROJECT LEAD. THERE WILL BE A FIRST PRIORITY, SECOND PRIORITY, AND THIRD PRIORITY. IF RELATIONSHIPS DIFFER IN INTERVIEW THAN WHAT WAS COLLECTED IN SCREENING, PRIORITIZE THOSE THAT ARE EITHER UNRELATED OR HAS A DISTANT RELATIONSHIP TO THE PARTICIPANT. THIS CAN INCLUDE GRANDPARENT/GRANDCHILD, AUNT/UNCLE/NEPHEW/NIECE, COUSIN, ETC.]

SECTION V: COGNITIVE INTERVIEW

[BRING UP POWERPOINT OF SURVEY QUESTIONS. SHOW SURVEY QUESTIONS FOR THE PROBES RELATED TO EACH QUESTION.]

Thank you for answering the questions. Now I'd like to ask you some questions about the answers you just provided.

GENERAL PROBES FOR SELF-ADMINISTERED WEB

In general, was this survey easy or difficult to complete? Please tell me more about that.

This survey that you completed currently asks questions via face to face or telephone. Do you think it would have been easier to answer questions via face-to-face/telephone or online like you just completed? Please tell me more about that.

[IF MULTIPLE MEMBERS IN HOUSEHOLD] There was a series of questions where you were asked to select who in the household participated in activities using the Internet. For example, some of these questions included whether household members use a laptop or smartphone or household members use the Internet at coffee shop or library. Was answering these questions on behalf of the entire household easy or difficult? **[IF NEEDED]** Can you tell me more about that?

Were there any questions where it was easier to answer for one person in the household than another? **(IF YES)** Can you tell me more about that?

PROBES FOR INDIVIDUAL QUESTIONS IN SELF-ADMINISTERED

First, we asked about the computers and other Internet-connected devices that members of your household have used at least occasionally during the past 6 months.

DESKTOP

We asked if anyone in this household uses a desktop computer at least occasionally.

In your own words, what does “desktop computer” mean to you in this question?

I am now showing the full question wording. Please focus on the wording that describes what a desktop is. Does this definition match with how you think of a desktop computer? **(IF NO)** Can you please tell me how you would describe a desktop?

(IF YES TO LAPTOP OR DESKTOP) Can you describe the type of computer(s) that your household has used?

TABLET

Earlier we asked if your household had a tablet or e-book reader.

(IF ANY DIFFICULTY) Can you tell me more about your answer to this question?

In your own words, what is the difference between a laptop and tablet, if any?

(IF YES TO LAPTOP AND DIDN'T EXPLAIN LAPTOP BEFORE) Can you tell me about the laptop computer(s) your household has used?

(IF YES to TABLET) Can you tell me more about the “tablet(s)” your household has used?

WEARAB

Earlier we asked if anyone in your household uses a wearable device that is connected to the Internet, such as a smart watch or fitness band. Was this question easy or difficult to answer?

Can you tell me more about your answer to this question?

Are there any other types of devices you consider to be “wearables,” other than smart watches and fitness bands?

TVBOX

Earlier we asked if anyone in the household uses a smart TV, a game system, or another device that connects to the Internet and plays through a TV.

Was this question easy or difficult to answer? [IF DIFFICULT] Can you tell me more about difficulty you experienced answering this question?

Can you tell me more about your answer to this question?

[IF TIME PERMITS] Other than Xbox, Apple TV, or Roku, are there any other devices that you thought about including but didn't?

DEVQUA

Earlier, we asked, how well do the computers and other Internet-connected devices used by this household work overall?

Was this question easy or difficult to answer?

Tell me more about your answer to this question. Can you tell me more about any times when computers and devices were not working well or perfectly? What happened? [IF NEEDED]

Were you thinking strictly about any external factors that may affect device usage, such as weather events or local power outages?

This question asked how well computers and other Internet-connected devices work overall.

What does the phrase "work poorly" mean to you in this question?

DEVSTA (ONLY IF TIME PERMITS)

Earlier we asked was anyone unable to use a computer or other Internet-connected device because it was in use by someone else, located elsewhere, or otherwise inaccessible?

Was this question easy or difficult to answer? [IF NEEDED] Can you tell me more about the difficulty you experienced answering this question?

Can you tell me more about your answer to this question?

INTRAV (ONLY IF TIME PERMITS)

Next, we asked questions about where [you/members of your household] currently use the Internet.

We asked if [you/anyone in the household] use(s) the Internet while going from place to place.

Was this question easy or difficult to answer? (IF NEEDED) Can you please tell me more about the difficulty you experienced answering this question?

(IF NEEDED) Can you tell me more about your answer to this question?

(IF NEEDED) Other than using Internet on public transit or using Google Maps/Waze, were there any other ways of using the Internet from place to place you thought of when answering this question?

INHOME to INOTHR General Probes

Earlier, you answered a series of questions about [you/your household's] Internet use at different locations including at home, work, school, coffee shops, etc. Which devices were [you/your household] thinking of as you answered these questions?

MOBDAT (ONLY IF TIME PERMITS)

Earlier, we asked if anyone in the household accessed the Internet using a data plan for a cell phone, smartphone, tablet, mobile hotspot, or other device. Was this question easy or difficult to answer?

Can you tell me more about your answer?

HOMTE

Earlier we asked you about the ways people access the Internet from their homes. You answered [INSERT ANSWER]. Can you tell me more about your answer to this question?

[IF YES TO 'SOME OTHER SERVICE' IN HOMTE]. I also asked if you had "some other service" and you said [INSERT ANSWER FROM HOMTE/TEOTHR]. Can you tell me more about this "other service"? Can you think of any other examples that would fit under "other service"?

[IF NO TO SOME OTHER SERVICE IN HOMTE] Can you think of any other types of ways that people can access the Internet?

Now I'm going to show you another question. Please answer and remember to think aloud:

Who is your home Internet Service Provider?

- Xfinity
- Spectrum
- AT&T
- Verizon
- Cox
- Altice
- CenturyLink or Quantum Fiber
- Frontier
- T-Mobile
- Other-please specify:

[IF AT&T]: How does AT&T provide the Internet service you use at home?

Select all that apply.

- A wire runs into my home and is connected to a modem or router.
- My/our mobile phone(s) or other devices connect directly to AT&T's wireless network (not via WiFi), just like outside of home.

[IF Verizon]: How does Verizon provide the Internet service you use at home?

Select all that apply.

- A wire runs into my home and is connected to a modem or router.
- My/our mobile phone(s) or other devices connect directly to Verizon's wireless network (not via WiFi), just like outside of home.
- Verizon provided me/us with a device that connects to their wireless network and creates a WiFi network for my/our home.

[IF T-Mobile]: How does T-Mobile provide the Internet service you use at home?

Select all that apply.

- My/our mobile phone(s) or other devices connect directly to T-Mobile's wireless network (not via WiFi), just like outside of home.
- T-Mobile provided me/us with a device that connects to their wireless network and creates a WiFi network for my/our home.

Were these questions answering about your Internet Service Provider easy or difficult? [IF NEEDED] Can you tell me more about that?

[IF ANSWERED ONE OF FOLLOW UP QUESTIONS] Can you tell me more about your answers for how AT&T/Verizon/T-Mobile provide your Internet service?

[IF THEY SAID “A WIRE RUNS INTO MY HOME...”] Is your Internet service either Fios or DSL?

[IF THEY SAID “PROVIDED ME/US WITH A DEVICE...”] You mentioned you were provided a device that connects to the wireless network. Is this like a Home 5G network?

[IF THEY DID NOT REPORT “MOBILE PHONE CONNECTS DIRECTLY to WIRELESS NETWORK” AND IT SEEMS LIKE THEY SHOULD HAVE] Does your Internet service provider allow your mobile phone to connect to their wireless network outside of the home?

[SHOW BOTH ORIGINAL SURVEY QUESTION AND ALTERNATE QUESTION] Now I’m going to show you both questions. Was it easier to answer one question or the other? Tell me more about that.

EMAIL to MEDMON Proxy Reporting (IF NEEDED)

Earlier, you answered a series of questions about [INSERT NAME]’s Internet use. These questions covered a variety of Internet use activities like checking email, watching videos, working, providing services, etc. I will show you these questions on screen.

How easy or difficult was it to answer these various Internet use questions about [INSERT NAME]?

[IF NEEDED] Which questions were easy to answer? [IF NEEDED] Can you tell me more about that?

[IF NEEDED] Which questions were difficult to answer? [IF NEEDED] Can you tell me more about that?

CONFER

Earlier we asked if [INSERT NAME] participated in video or voice calls or conferencing over the Internet. Was this question easy or difficult to answer? Can you tell me more about your answer?

[IF NEEDED] Have [INSERT NAME] used WhatsApp for any video or voice calls in the past 6 months?

VIDEO AND AUDIO

Earlier we asked questions about if [INSERT NAME] watched videos or listened to music, radio programs, and podcasts over the Internet.

Would these questions have been easier, more difficult to answer, or about the same if we had used “streaming or downloading” instead of “watching or listening”? Do these phrases mean the same thing to you or do they mean something different?

ESRVCS

Earlier we asked if [INSERT NAME] offered their own services for sale via the Internet.

Was this question easy or difficult to answer? Can you tell me more about your answer to this question?

FINANC

Earlier we asked if [INSERT NAME] used the Internet for financial services such as banking, investing, paying bills online, or sending money to other people.

Was this question easy or difficult to answer? Can you tell me more about your answer to this question?

HOMIOT

Earlier we asked if [INSERT NAME] interacted with household equipment or appliances that are connected to the Internet, such as a smart thermostat, light bulb, or security system. Was this question easy or difficult to answer?

IF NEEDED Can you tell me more about your answer?

Some of the examples you see here provided in this question included ‘Nest Thermostat’ and ‘Ring Camera’. Were these examples helpful to you when answering? [IF NEEDED] Are you familiar with the ‘Nest Thermostat’ and/or ‘Ring Camera’?

MEDINT

Earlier, we asked a series of questions about whether your household used the Internet for health-related activities during the past six months. These questions asked about access to health records, communicating with doctors, researching health information online, and whether an electronic health monitoring service was used.

Did you answer these questions thinking about your household in general or just about your own experience?

(IF ANSWERING ON BEHALF OF HOUSEHOLD) How easy or difficult was it to answer about the household?

(FOR EVERYONE) Can you tell me more about your answers to these questions?

PSPRE

- (1) Conducting financial transactions such as banking, investing, or paying bills online?
- (2) Buying goods or services online?
- (3) Posting photos, updates, or other information on social networks?
- (4) Expressing an opinion on a controversial or political issue, such as on a blog or social network, or in an email?
- (5) Finding information using Google Search, Microsoft Bing, or another search engine?

Next, we asked if anyone in the household had concerns about privacy or security that stopped them from doing specific activities online.

What time frame were you thinking about when you answered this question?

Was this question easy or difficult? Can you tell me more about that?

Can you tell me more about your answer(s) to this question?

[IF DIFFICULTY ANSWERING] I noticed that you hesitated when answering **[READ OPTION FROM LIST]**. Can you tell me more about what you were thinking as you answered this question?

PSCON

Next, we asked what concerns household members the most about online privacy or security risks.

Was this question easy or difficult? Can you tell me more about that?

Can you tell me more about the answer(s) you selected to this question?

Are there any options on this list that you find confusing? **[IF YES]**. Can you tell me which options?

[IF ONLY ONE SELECTED] Did you notice you can select more than one option for this question?

NOHM (IF QUESTION WAS ASKED IN INTERVIEW)

Next, we asked what concerns household members the most about online privacy or security risks.

Was this question easy or difficult? Can you tell me more about that?

Can you tell me more about your answer(s) you selected to this question?

Are there any options on this list that you find confusing? [IF YES]. Can you tell me which options?

[IF ONLY ONE SELECTED] Did you notice you can select more than one option for this question?

PROBES FOR INDIVIDUAL QUESTIONS NOT ASKED ABOUT YET

I noticed you commented on (INSERT QUESTION). Can you tell me more about what you were thinking?

I noticed you hesitated when answering (INSERT QUESTION). Can you tell me more about what were you thinking?

SECTION VI: Debriefing Questions

Overall, how easy or difficult was it to answer questions for your household?

Do you think other people might find any of these questions sensitive?

- [IF YES] Which ones?
- [IF NEEDED] What about the question do you think people would find sensitive?

Is there anything else you would like to tell us that you haven't had a chance to mention yet?

Those are all the questions I have for you. Thank you for participating in this study. Your feedback has been very helpful.

TURN OFF RECORDER

SECTION VII: COMPENSATION FOR PARTICIPATION

IF SIGNED CONSENT/VOUCHER BEFORE INTERVIEW

We will be mailing out that \$40 cash incentive in the next few days.

IF DID NOT SIGN CONSENT/VOUCHER FORM BEFORE INTERVIEW

Just as a reminder, you need to sign the document in the link before we can send the \$40 cash incentive. Would you like me to send the link again and/or talk you through any problems you may be having with it?

[IF SEND AGAIN] Just to confirm, what e-mail address would you like me to send that to?

Thank you very much for your time today and we will be mailing out that \$40 cash incentive as soon as you sign that document.

CLOSE OUT STEPS

CHECK RECORDING IS THE RIGHT LENGTH AND SAVE TO THE PARTICIPANT'S FOLDER.

MAKE A NOTE OF ANY TECHNOLOGICAL ISSUES YOU HAD DURING THE INTERVIEW BELOW:

Appendix B. Self-Administered NTIA Internet Use Survey (Single Person Households)

CMPINT
All households

This month we are asking some additional questions about Internet use. First, we will ask what types of computers and other devices you currently use. Please focus on devices you currently have access to, and have used at least occasionally during the past six months. We are interested in devices used at home, work, school, a library, or any other place.

LAPTOP
All households

Do you use a laptop computer at least occasionally? Include any laptop computers you currently have access to and have used during the past six months.

A laptop computer is a personal computer that is portable and includes a built-in keyboard and screen.

- (1) Yes
- (2) No

DESKTP
All households

Do you use a desktop computer?

A desktop computer is a personal computer that must be plugged into a power outlet and is typically at a single location.

- (1) Yes
- (2) No

TABLET
All households

Do you use a tablet or e-book reader, such as an iPad or Kindle?

A tablet or e-book reader is a personal computer that is portable and has a built-in screen, but does not have a built-in physical keyboard.

- (1) Yes
- (2) No

MPHONE
All households

Do you use a smartphone, or a cell phone that connects to the Internet?

- (1) Yes
- (2) No

WEARAB Do you use a wearable device that is connected to the Internet, such as a smart watch or fitness band?
All households
Examples include an Apple Watch or Fitbit.

- (1) Yes
- (2) No

TVBOX Do you use a smart TV, a game system, or another device that connects to the Internet and plays through a TV?
All households
Examples include an Apple TV, Xbox, or Roku that can access the Internet.

- (1) Yes
- (2) No

If any (DESKTP LAPTOP TABLET MPHONE WEARAB TVBOX)[1-16] == 1 (anyone in the household uses any device), go to DEVQUA
Else go to INTINT

DEVQUA Thinking about all the different devices you use, how well do the computers and other Internet-connected devices work overall?
Households w/
any device use
Please do not include home Wi-Fi routers or similar equipment.

- (1) They do not work at all
- (2) They work poorly
- (3) They somewhat work
- (4) They work well
- (5) They work perfectly

DEVSTA At any point during the past six months, were you unable to use a computer or other Internet-connected device because it was in use by someone else, located elsewhere, or otherwise inaccessible?
Households w/
any device use

- (1) Yes
- (2) No

INTINT Next, we will ask where you currently use the Internet. Please focus on places where you have used the Internet at least occasionally during the past six months. Internet use includes a wide variety of activities on all devices, from checking email or browsing the Web to watching videos or using mobile apps.
All households

INHOME Do you use the Internet at home?
All households *This includes accessing the Internet with a cell phone, computer, tablet, or other device.*

- (1) Yes
- (2) No

**If any PEMLR[1-16] == 1 or 2 (anyone in the household is employed), go to INWORK
Else go to INSCHL**

INWORK Do you use the Internet at work?
HHs w/ at least 1 employed person *Do not include work from home.*

- (1) Yes
- (2) No

INSCHL Do you use the Internet at school?
All households *Please only include use at school as a student. Do not include taking online classes from a location other than a school campus.*

- (1) Yes
- (2) No

INCAFE Do you use the Internet at a coffee shop or other business that offers Internet access?
All households

- (1) Yes
- (2) No

In the past six months...

INTRAV Do you use the Internet while going from place to place?
All households *Examples include using the Internet while riding public transit, or using Google Maps or Waze on your cell phone while driving.*

- (1) Yes
- (2) No

INLICO Do you use the Internet at a library, community center, park, or other public place?
All households

- (1) Yes
- (2) No

INELHO Do you use the Internet at someone else's home?
All households

- (1) Yes
- (2) No

INOTHR Do you use the Internet at some other location we haven't covered?
All households

- (1) Yes
- (2) No

If any IN*[1-16] == 1 (anyone goes online anywhere), go to TCHINT
Else go to NONINT

TCHINT Next, we are interested in learning about the specific technologies households
HHs w/ any use to go online. Please focus on how your household as a whole connects to
Internet users, any the Internet.
location

MOBDAT Do you access the Internet using a data plan for a cell phone, smartphone,
HHs w/ any tablet, mobile hotspot, or other device?
Internet users, any *This type of Internet service is provided by a wireless carrier, and may be part of*
location *a package that also includes voice calls from a cell phone or smartphone.*

- (1) Yes
- (2) No

If any INHOME[1-16] == 1 & MOBDAT == 1 (anyone goes online at home and household has a mobile data plan), go to HMINT1
Else if INHOME[1-16] == 1 & MOBDAT == 2 (anyone goes online at home but household does not have a mobile data plan), go to HMINT2
Else go to USEINT

HMINT1 In addition to your mobile Internet service or data plan, we are interested in
HHs w/ any home whether you also use any other type of Internet service when at home.
Internet users &
MOBDAT == 1

Go to HOMTE

HMINT2 You mentioned that you use the Internet at home.
HHs w/ any home
Internet users &
MOBDAT == 2

HOMTE The following questions ask the different ways people access the Internet from
HHs w/ any home Internet users their homes, **other than a mobile data plan**. At home, do you access the Internet using:

- 1) High-speed Internet service installed at home, such as cable, DSL, or fiber-optic service?
This type of Internet service is often provided by a cable company or phone company.

- (1) Yes
- (2) No

- 2) Satellite Internet service?
This type of Internet service is received through a satellite dish installed at home.

- (1) Yes
- (2) No

- 3) Some other service?

- (1) Yes (**Go to TEOTHR**)
- (2) No (**Go to HNETQL**)

TEOTHR What is the other service?
HHs w/ any home Internet users & [open ended text box]
HOMTE4 == 1

HNETQL In your experience over the past six months, has your Internet connection at
HHs w/ any home Internet users home met your needs, including for speed, reliability, and, if applicable, data caps?

- (1) Yes
- (2) No

HNETST At any point during the past six months, did you temporarily lose a home
HHs w/ any home Internet users Internet connection due to difficulty paying?

- (1) Yes
- (2) No

Random age 15+ Internet user in household (any IN*[1-16] or NETCHK == 1 & PRTAGE >= 15) selected for questions EMAIL-HOMIOT.

If all Internet users are under 15 years old, skip to MEDINT.

USEINT Random respondents	We are interested in learning about the applications and services people use on the Internet. Consider how you personally currently use the technologies you have reported, whether at home or any other location. Please focus on activities you have done online at least occasionally during the past six months.
EMAIL Random respondents	Do you use email? (1) Yes (2) No
TEXTIM Random respondents	Do you text or use instant messaging? (1) Yes (2) No
SOCIAL Random respondents	Do you use social networks, such as Facebook, Twitter, or Instagram? (1) Yes (2) No In the past six months...
GAMING Random respondents	Have you played video games online, whether on a smartphone, console, PC, or any other computing device? (1) Yes (2) No
CONFER Random respondents	Do you participate in video or voice calls or conferencing over the Internet, such as with FaceTime, Zoom, or WhatsApp? (1) Yes (2) No
VIDEO Random respondents	Do you watch videos over the Internet? <i>Examples include streaming or downloading videos on YouTube and Netflix.</i> (1) Yes (2) No In the past six months...

AUDIO
Random
respondents

Have you listened to music, radio programs, or podcasts using the Internet?

(1) Yes
(2) No

PUBLSH
Random
respondents

Do you post or upload blog posts, original videos, or other content to the Internet that you created?

Do not include photos or videos taken for personal use and shared to social media.

(1) Yes
(2) No

If PEMLR[selected respondent] == 1 or 2 (respondent is employed), go to TELEWK
Else go to JOBSCH

TELEWK
Employed
random
respondents

Do you use the Internet to telecommute or work while away from your usual workplace, such as working from home?

(1) Yes
(2) No

In the past six months...

JOBSCH
Random
respondents

Have you used the Internet to search or apply for a job?

(1) Yes
(2) No

EDTRAI
Random
respondents

Do you use the Internet for online classes or job training?

(1) Yes
(2) No

EGOVTS
Random
respondents

Do you use the Internet to access government services, such as registering to vote or renewing your driver's license?

(1) Yes
(2) No

In the past six months...

USESVC Random respondents	Have you used the Internet to request services provided by other people? <i>Examples include hailing a ride using Uber, ordering food using DoorDash or Instacart, reserving a room through Airbnb, or requesting a home repair through Angi.</i> (1) Yes (2) No
ESRVCS Random respondents	Do you offer your own services for sale via the Internet? <i>Examples include offering rentals on Airbnb and driving for Uber or Lyft. Do not include any goods or possessions sold online, such as clothing, shoes, or crafts.</i> (1) Yes (2) No
ECOMME Random respondents	Do you use the Internet for online shopping, travel reservations, or other consumer services? (1) Yes (2) No
EGOODS Random respondents	Do you use the Internet to sell goods? <i>Examples include selling items on Etsy or eBay.</i> (1) Yes (2) No In the past six months...
FINANC Random respondents	Have you used the Internet for financial services such as online banking, investing, paying bills, or sending money to other people using services like CashApp, Venmo, or PayPal? <i>Examples include using your bank's website or app, and paying a credit card or utility bill online.</i> (1) Yes (2) No
VOICEA Random respondents	Do you communicate with a digital assistant, such as Alexa, Siri, or Google Assistant? (1) Yes (2) No

HOMIOT Random respondents	Do you interact with household equipment or appliances that are connected to the Internet, such as a smart thermostat, light bulb, or security system? <i>Examples include nest thermostat and ring camera.</i>
	(1) Yes (2) No
MEDINT HHs w/ any Internet users, any location	The following questions ask about how you have used the Internet for health-related activities during the past six months.
MEDREC HHs w/ any Internet users, any location	Do you access health records or health insurance records online?
	(1) Yes (2) No
MEDDOC HHs w/ any Internet users, any location	Do you communicate with a doctor or other health professional using the Internet? <i>Examples include telehealth appointments and communicating over a patient portal or email.</i>
	(1) Yes (2) No
MEDINF HHs w/ any Internet users, any location	Do you research health information online, such as with WebMD or similar services?
	(1) Yes (2) No
MEDMON HHs w/ any Internet users, any location	Do you use an electronic health monitoring service that collects and sends data to your doctor or health care provider through the Internet? <i>Examples include connected devices that monitor vital statistics, blood glucose levels, or blood pressure.</i>
	(1) Yes (2) No
UHHINT HHs w/ any Internet users, any location	We are interested in learning what challenges you have had while using the Internet.

PSPRE

HHs w/ any
Internet users, any
location

During the past 12 months, have concerns about privacy or security stopped you from:

a. Conducting financial transactions such as banking, investing, or paying bills online?

(1) Yes

(2) No

b. Buying goods or services online?

(1) Yes

(2) No

c. Posting photos or other information on social networks?

(1) Yes

(2) No

d. Expressing an opinion on a controversial or political issue, such as on a blog or social network, or in an email?

(1) Yes

(2) No

e. Finding information using Google or another search engine?

(1) Yes

(2) No

PSCON

HHs w/ any
Internet users, any
location

Overall, what concerns you the most when it comes to online privacy and security risks?

Select all that apply.

(1) Identity theft

(2) Credit card or banking fraud

(3) Data collection or tracking by online services

(4) Data collection or tracking by government

(5) Loss of control over personal data such as email or social network profiles

(6) Threats to personal safety, such as online harassment, stalking, or cyber-bullying

(7) No concerns

(8) Other- please specify: _____

PSCYBA During the past 12 months, have you experienced an online security breach,
HHs w/ any identity theft, or a similar crime?
Internet users, any
location
(1) Yes
(2) No

If any INHOME[1-16] == 1 (anyone goes online at home), go to TVINT
Else go to NHMINT

NHMINT You previously mentioned that you use the Internet in some places, but not at
HHs w/ any home.
Internet users, but
none from home

Go to EVRHOM

NONINT
HHs w/o any For our remaining questions, we are interested in learning why you do not use
Internet users the Internet.
from anywhere

EVRRHOM Have you ever used the Internet from home?
HHs w/o any (1) Yes
home Internet (2) No
users

NOHM What are the reasons why you do not use the Internet at home?
HHs w/o any *Select all that apply.*
home Internet (1) Don't need it or not interested
users (2) Can't afford it
(3) Not worth the cost
(4) Can use it elsewhere
(5) Not available in area
(6) No computing device, or device inadequate or broken
(7) Online privacy or cybersecurity concerns
(8) Personal safety concerns
(9) Household moved or is in the process of moving
(10) Other- please specify: _____

If more than one NOHM response given, go to PRINOH
Else go to LOPRCE

PRINOH HHs w/o any home Internet users & more than one NOHM response	Of the reasons you just listed for not going online at home, which do you consider to be the most important? <i>[Autofill selected options from NOHM]</i>
LOPRCE HHs w/o any home Internet users	At what monthly price, if any, would you buy home Internet service? <i>Enter number in dollars or zero if you would not buy it.</i>
TVINT All households	Our last few questions are about the use of cable TV and satellite TV services not accessed through the Internet.
TRADTV All households	Do you subscribe to cable TV or satellite TV? <i>This often requires a cable box or satellite dish. Do not include TV streaming subscriptions, such as Netflix, Sling TV, or Hulu.</i>
	(1) Yes (End Interview) (2) No (Go to PREVTv)
PREVTv HHs w/o any cable TV subscribers	Have you ever subscribed to cable TV or satellite TV? <i>Do not include TV streaming subscriptions, such as Netflix, Sling TV, or Hulu.</i>
	(1) Yes (2) No

Appendix C. Self-Administered NTIA Internet Use Survey (Multiple Person Households)

CMPINT

All households

This month we are asking some additional questions about Internet use. First, we will ask what types of computers and other devices members of your household currently use. You will be asked who in your household uses each device. Please focus on devices members of your household currently have access to, and have used at least occasionally during the past six months. We are interested in devices used at home, work, school, a library, or any other place.

LAPTOP

All households

Who in your household uses a laptop computer at least occasionally? Include any laptop computers members of your household currently have access to and have used during the past six months.

A laptop computer is a personal computer that is portable and includes a built-in keyboard and screen.

Select all that apply.

Members of household are listed vertically (1-16)

No one in this household

DESKTP

All households

Who uses a desktop computer?

A desktop computer is a personal computer that must be plugged into a power outlet and is typically at a single location.

Select all that apply.

Members of household are listed vertically (1-16)

No one in this household

TABLET

All households

Who uses a tablet or e-book reader, such as an iPad or Kindle?

A tablet or e-book reader is a personal computer that is portable and has a built-in screen, but does not have a built-in physical keyboard.

Select all that apply.

Members of household are listed vertically (1-16)

No one in this household

In the past six months...

MPHONE Who uses a smartphone, or a cell phone that connects to the Internet?
All households *Select all that apply.*

Members of household are listed vertically (1-16)
No one in this household

WEARAB Who uses a wearable device that is connected to the Internet, such as a smart watch or fitness band?
All households *Examples include an Apple Watch or Fitbit.*
Select all that apply.

Members of household are listed vertically (1-16)
No one in this household

TVBOX Who uses a smart TV, a game system, or another device that connects to the Internet and plays through a TV?
All households *Examples include an Apple TV, Xbox, or Roku that can access the Internet.*
Select all that apply.

Members of household are listed vertically (1-16)
No one in this household

If any (DESKTOP LAPTOP TABLET MPHONE WEARAB TVBOX)[1-16] == 1 (anyone in the household uses any device), go to DEVQUA
Else go to INTINT

DEVQUA Thinking about all the different devices used by this household, how well do the computers and other Internet-connected devices work overall?
Households w/ any device use *Please do not include home Wi-Fi routers or similar equipment.*

- (1) They do not work at all
- (2) They work poorly
- (3) They somewhat work
- (4) They work well
- (5) They work perfectly

DEVSTA
Households w/
any device use

At any point during the past six months, were members of your household unable to use a computer or other Internet-connected device because it was in use by someone else, located elsewhere, or otherwise inaccessible?

(1) Yes
(2) No

INTINT
All households

Next, we will ask where members of your household currently use the Internet. You will be asked who in your household uses the Internet at each location. Please focus on places where members of your household have used the Internet at least occasionally during the past six months. Internet use includes a wide variety of activities on all devices, from checking email or browsing the Web to watching videos or using mobile apps.

INHOME
All households

Who in your household uses the Internet at home?
This includes accessing the Internet with a cell phone, computer, tablet, or other device.
Select all that apply.

Members of household are listed vertically (1-16)
No one in this household

If any PEMLR[1-16] == 1 or 2 (anyone in the household is employed), go to INWORK
Else go to INSCHL

INWORK
HHs w/ at least
1 employed
person

Who uses the Internet at work?
Do not include work from home.
Select all that apply.

Members of household are listed vertically (1-16)
No one in this household

INSCHL
All households

Who uses the Internet at school?
Please count students only. Do not include taking online classes from a location other than a school campus.
Select all that apply.

Members of household are listed vertically (1-16)
No one in this household

INCAFE Who uses the Internet at a coffee shop or other business that offers Internet access?
All households *Select all that apply.*

Members of household are listed vertically (1-16)
No one in this household

In the past six months...

INTRAV Who uses the Internet while going from place to place?
All households *Examples include using the Internet while riding public transit, or using Google Maps or Waze on your cell phone while driving.*
Select all that apply.

Members of household are listed vertically (1-16)
No one in this household

INLICO Who uses the Internet at a library, community center, park, or other public place?
All households *Select all that apply.*

Members of household are listed vertically (1-16)
No one in this household

INELHO Who uses the Internet at someone else's home?
All households *Select all that apply.*

Members of household are listed vertically (1-16)
No one in this household

INOTHR Who uses the Internet at some other location we haven't covered?
All households *Select all that apply.*

Members of household are listed vertically (1-16)
No one in this household

If HRNUMHOU > 1 & (total persons where any IN* == 1) < HRNUMHOU (this is a multi-person household and not everyone was reported as using the Internet), go to NETCHK
Else if any IN*[1-16] == 1 (anyone goes online anywhere), go to TCHINT
Else go to NONINT

NETCHK

Multi-person
HHs reporting
any non-
Internet users

Based on your responses, it sounds like *(list names where all IN* != 1)*
[does/do] not use the Internet. [Does/Do] [he/she/you/these people in your
household] use the Internet from any location, and for any purpose?

(1) Yes

(2) No

*(If additional users) Who uses the Internet from any location, and for any
purpose?*

Fill appropriate lines if necessary; locations of use will be allocated

If any IN*[1-16] or NETCHK == 1 (anyone goes online anywhere), go to TCHINT

Else go to NONINT

TCHINT

HHs w/ any
Internet users,
any location

Next, we are interested in learning about the specific technologies households
use to go online. Please focus on how your household as a whole connects to
the Internet.

MOBDAT

HHs w/ any
Internet users,
any location

Does anyone in this household access the Internet using a data plan for a cell
phone, smartphone, tablet, mobile hotspot, or other device?

*This type of Internet service is provided by a wireless carrier, and may be part
of a package that also includes voice calls from a cell phone or smartphone.*

(1) Yes

(2) No

**If any INHOME[1-16] == 1 & MOBDAT == 1 (anyone goes online at home and household has
a mobile data plan), go to HMINT1**

**Else if INHOME[1-16] == 1 & MOBDAT == 2 (anyone goes online at home but household
does not have a mobile data plan), go to HMINT2**

Else go to USEINT

HMINT1

HHs w/ any
home Internet
users &
MOBDAT ==
1

In addition to your household's mobile Internet service or data plan, we are
interested in whether your household also uses any other type of Internet
service when at home.

Go to HOMTE

HMINT2

You mentioned that members of this household use the Internet at home.

HHs w/ any
home Internet
users &
MOBDAT ==
2

HOMTE

HHs w/ any
home Internet
users

The following questions ask the different ways people access the Internet from their homes, **other than a mobile data plan**. At home, does anyone in this household access the Internet using:

- 1) High-speed Internet service installed at home, such as cable, DSL, or fiber-optic service?

This type of Internet service is often provided by a cable company or phone company.

(1) Yes

(2) No

- 2) Satellite Internet service?

This type of Internet service is received through a satellite dish installed at home.

(1) Yes

(2) No

- 3) Some other service?

(1) Yes **(Go to TEOTHR)**

(2) No **(Go to HNETQL)**

TEOTHR

HHs w/ any
home Internet
users &
HOMTE4 == 1

What is the other service?

[open ended text box]

HNETQL

HHs w/ any
home Internet
users

In your household's experience over the past six months, has your household's Internet connection at home met the household's needs, including for speed, reliability, and, if applicable, data caps?

(1) Yes

(2) No

HNETST

HHs w/ any
home Internet
users

At any point during the past six months, did this household temporarily lose a home Internet connection due to difficulty paying?

(1) Yes

(2) No

Random age 15+ Internet user in household (any IN*[1-16] or NETCHK == 1 & PRTAGE >= 15) selected for questions EMAIL-HOMIOT.

If all Internet users are under 15 years old, skip to MEDINT.

USEINT

Random
respondents

We are interested in learning about the applications and services people use on the Internet. We have randomly selected one member of your household for this section. Consider how [you personally/NAME] currently use[s] the technologies you have reported, whether at home or any other location. Please focus on activities [you have/NAME has] done online at least occasionally during the past six months.

EMAIL

Random
respondents

[Do you/Does NAME] use email?

(1) Yes

(2) No

TEXTIM

Random
respondents

[Do you/Does NAME] text or use instant messaging?

(1) Yes

(2) No

SOCIAL

Random
respondents

[Do you/Does NAME] use social networks, such as Facebook, Twitter, or Instagram?

(1) Yes

(2) No

In the past six months...

GAMING

Random
respondents

[Have you/Has NAME] played video games online, whether on a smartphone, console, PC, or any other computing device?

(1) Yes

(2) No

CONFER

Random
respondents

[Do you/Does NAME] participate in video or voice calls or conferencing over the Internet, such as with FaceTime, Zoom, or WhatsApp?

(1) Yes

(2) No

VIDEO

Random
respondents

[Do you/Does NAME] watch videos over the Internet?

Examples include streaming or downloading videos on YouTube and Netflix.

(1) Yes

(2) No

In the past six months...

AUDIO

Random
respondents

[Have you/Has NAME] listen to music, radio programs, or podcasts using the Internet?

(1) Yes

(2) No

PUBLISH

Random
respondents

[Do you/Does NAME] post or upload blog posts, original videos, or other content to the Internet that [you/NAME] created?

Do not include photos or videos taken for personal use and shared to social media.

(1) Yes

(2) No

If PEMLR[selected respondent] == 1 or 2 (respondent is employed), go to TELEWK

Else go to JOBSCH

TELEWK

Employed
random
respondents

[Do you/Does NAME] use the Internet to telecommute or work while away from [your/his/her] usual workplace, such as working from home?

- (1) Yes
- (2) No

In the past six months...

JOBSCH

Random
respondents

[Have you/Has NAME] used the Internet to search or apply for a job?

- (1) Yes
- (2) No

EDTRAI

Random
respondents

[Do you/Does NAME] use the Internet for online classes or job training?

- (1) Yes
- (2) No

EGOVTS

Random
respondents

[Do you/Does NAME] use the Internet to access government services, such as registering to vote or renewing your driver's license?

- (1) Yes
- (2) No

In the past six months...

USESVC

Random
respondents

[Have you/Has NAME] used the Internet to request services provided by other people?

Examples include hailing a ride using Uber, ordering food using DoorDash or Instacart, reserving a room through Airbnb, or requesting a home repair through Angi.

- (1) Yes
- (2) No

ESRVCS Random respondents	[Do you/Does NAME] offer [your/his/her] own services for sale via the Internet? <i>Examples include offering rentals on Airbnb and driving for Uber or Lyft. Do not include any goods or possessions sold online, such as clothing, shoes, or crafts.</i> (1) Yes (2) No
ECOMME Random respondents	[Do you/Does NAME] use the Internet for online shopping, travel reservations, or other consumer services? (1) Yes (2) No
EGOODS Random respondents	[Do you/Does NAME] use the Internet to sell goods? <i>Examples include selling items on Etsy or eBay.</i> (1) Yes (2) No
FINANC Random respondents	In the past six months... [Have you/Has NAME] used the Internet for financial services such as online banking, investing, paying bills, or sending money to other people using services like CashApp, Venmo, or PayPal? <i>Examples include using your bank's website or app, and paying a credit card or utility bill online.</i> (1) Yes (2) No
VOICEA Random respondents	[Do you/Does NAME] communicate with a digital assistant, such as Alexa, Siri, or Google Assistant? (1) Yes (2) No
HOMIOT Random respondents	[Do you/Does NAME] interact with household equipment or appliances that are connected to the Internet, such as a smart thermostat, light bulb, or security system?

Examples include nest thermostat and ring camera.

(1) Yes

(2) No

MEDINT

HHs w/ any
Internet users,
any location

Our remaining questions are about your household as a whole. The following questions ask about how your household has used the Internet for health-related activities during the past six months.

MEDREC

HHs w/ any
Internet users,
any location

Does anyone in this household, including you, access health records or health insurance records online?

(1) Yes

(2) No

MEDDOC

HHs w/ any
Internet users,
any location

Does anyone in this household communicate with a doctor or other health professional using the Internet?

Examples include telehealth appointments and communicating over a patient portal or email.

(1) Yes

(2) No

MEDINF

HHs w/ any
Internet users,
any location

Does anyone in this household research health information online, such as with WebMD or similar services?

(1) Yes

(2) No

MEDMON

HHs w/ any
Internet users,
any location

Does anyone in this household use an electronic health monitoring service that collects and sends data to your doctor or health care provider through the Internet?

Examples include connected devices that monitor vital statistics, blood glucose levels, or blood pressure.

(1) Yes

(2) No

UHHINT

We are interested in learning generally, without identifying specific people, what challenges your household has had while using the Internet.

HHs w/ any
Internet users,
any location

PSPRE

HHs w/ any
Internet users,
any location

During the past 12 months, have concerns about privacy or security stopped anyone in this household from:

a. Conducting financial transactions such as banking, investing, or paying bills online?

(1) Yes

(2) No

b. Buying goods or services online?

(1) Yes

(2) No

c. Posting photos or other information on social networks?

(1) Yes

(2) No

d. Expressing an opinion on a controversial or political issue, such as on a blog or social network, or in an email?

(1) Yes

(2) No

e. Finding information using Google or another search engine?

(1) Yes

(2) No

PSCON

HHs w/ any
Internet users,
any location

Overall, what concerns members of this household the most when it comes to online privacy and security risks?

Select all that apply

- (1) Identity theft
- (2) Credit card or banking fraud
- (3) Data collection or tracking by online services
- (4) Data collection or tracking by government
- (5) Loss of control over personal data such as email or social network profiles
- (6) Threats to personal safety, such as online harassment, stalking, or cyber-bullying
- (7) No concerns
- (8) Other – please specify: _____

PSCYBA

HHs w/ any
Internet users,
any location

During the 12 months, has any member of your household experienced an online security breach, identity theft, or a similar crime?

- (1) Yes
- (2) No

If any INHOME[1-16] == 1 (anyone goes online at home), go to TVINT

Else go to NHMINT

NHMINT

HHs w/ any
Internet users,
but none from
home

You previously mentioned that members of your household use the Internet in some places, but not at home.

Go to EVRHOM

NONINT

HHs w/o any
Internet users
from anywhere

Our remaining questions are about your household as a whole. We are interested in learning why households such as yours do not use the Internet.

EVRRHOM

HHs w/o any
home Internet
users

Has anyone in this household ever used the Internet from home?

- (1) Yes
- (2) No

NOHM

HHs w/o any
home Internet
users

What are the reasons why members of your household do not use the Internet at home?

Select all that apply.

- (1) Don't need it or not interested
- (2) Can't afford it
- (3) Not worth the cost
- (4) Can use it elsewhere
- (5) Not available in area
- (6) No computing device, or device inadequate or broken
- (7) Online privacy or cybersecurity concerns
- (8) Personal safety concerns
- (9) Household moved or is in the process of moving
- (10) Other- please specify: _____

If more than one NOHM response given, go to PRINOH

Else go to LOPRCE

PRINOH

HHs w/o any
home Internet
users & more
than one
NOHM
response

Of the reasons you just listed for not going online at home, which does your household consider to be the most important?

[Autofill selected options from NOHM]

LOPRCE

HHs w/o any
home Internet
users

At what monthly price, if any, would your household buy home Internet service?

Enter number in dollars or zero if you would not buy it.

TVINT

All households

Our last few questions are about the use of cable TV and satellite TV services not accessed through the Internet.

TRADTV

All households

Does anyone in this household subscribe to cable TV or satellite TV?

This often requires a cable box or satellite dish. Do not include TV streaming subscriptions, such as Netflix, Sling TV, or Hulu.

- (1) Yes **(End Interview)**
- (2) No **(Go to PREVT)**

PREVTV

HHs w/o any
cable TV
subscribers

Has anyone in this household ever subscribed to cable TV or satellite TV?

Do not include TV streaming subscriptions, such as Netflix, Sling TV, or Hulu.

(1) Yes

(2) No